



MOKO SMART



MK Sensor Series

APP User Manual

Version: v1.2

MOKO TECHNOLOGY LTD.

Revision History

Version	Date	Notes	Contributor(s)
V1.0	Dec 19, 2024	Initial version	Leo
V1.1	Jan 24, 2025	Modified APP description, MK Sensor series merged to BeaconX Pro APP	Leo
V1.2	May 20, 2025	1. Add product shutdown instruction 2. Delete Door Magnetic Related Functions	Leo

About this User Guide

This User Guide was designed to help users to know the overview of **MK Sensor** series products and set up **MOKO Sensor** through “**BeaconX Pro**” APP. Through this document, users will initial to know the product’s functional framework, the setup process of Beacon advertisement, connection, configuration, DFU update, etc.

This guide will not cover the sales administration and the ordering process. Some additional technical guides will be needed if further explanation is required.

- **Applicable version description:**

Android APP Version: “BeaconX Pro” V2.1.0 or above (“MK Sensor series” V1.0.0 or above)

iOS APP Version: “BeaconX Pro” V2.4.1 or above (“MK Sensor series” V1.0.0 or above)

Firmware Version: BXP-S Series V1.0.0 or above

- **Applicable products:**

Product model	ID picture
M4/ M4 Sensor/ M4 Pro	
L01 Series	
L02 Series	
M2 Sensor	
S05T	
H4-S/H4P-S	

NOTE:

The product series of MOKO may have different design variations for the same model. Please confirm that your device is the version compatible with the app.

Table of Contents

Revision History	2
About this User Guide	3
1. Overview	8
2. Content parsing of Beacon advertisement	9
2.1 Google Eddystone – UID	10
2.2 Google Eddystone – URL	11
2.3 Google Eddystone – TLM (Unencrypted)	12
2.4 Customized frame – “iBeacon”	13
2.5 Customized frame – “Sensor info”	15
3. APP user guidance	17
3.1 Discover your tag	17
3.1.1 Select the device type of your tag	17
3.1.2 Scanning your tag	18
3.1.3 Filtering your tag	18
3.2 Connect with your tag	19
3.3 Know your tag’s advertising logic	19
3.4 Access to SLOT configurations	20
3.4.1 Setting your tag’s advertisement SLOT	20
3.4.2 Setting your tag’s trigger function	21
3.4.3 Lock Event Occurs ADV Duration	24
3.5 How to apply to your scenarios	25
3.5.1 Environmental monitoring and analysis	25
3.5.2 Asset management	27
3.6 Access to sensor configurations	30
3.6.1 Motion sensor (3-axis accelerometer)	30
3.6.2 Temperature & Humidity sensor	31
3.7 Quick switch for basic features	32
3.7.1 Switch your Tag to non-connectable status	33
3.7.2 Disable trigger LED indicator	33
3.7.3 Disable password verification	33
3.7.4 Autofill Tag ID	33
3.7.5 Disable/Enable your Beacon’s button function	33

3.8 Factory reset your Beacon	33
3.8.1 Remote reset	33
3.8.2 Hardware reset	34
3.9 Modify “Connection password”	35
3.10 Remote reminder	35
3.11 Reset Battery	36
3.12 Battery ADV Mode	36
3.13 ADV Channel	37
3.14 How to enable “Long range” features	37
3.15 Upgrade your Tag’s firmware	39
3.15.1 Android – MOKO APP DFU instructions	39
3.15.2 iOS – MOKO APP DFU instructions	40
3.16 General information	41
3.17 Obtain log file from Android APP	41
3.18 How to power off your device	42
4. FAQ.....	43
Contact.....	44

List of Figures

Figure 1: Example content structure of Customized – iBeacon advertisement.....	14
Figure 2: Beacon firmware options interface preview(Android&IOS)	17
Figure 3: Example scanning preview of MK Sensor series	18
Figure 4: Configurations of scanning filter settings	18
Figure 5: Overall advertising logic by default parameters	19
Figure 6: Configurations of advertisement SLOT interface	20
Figure 7: Parameters and ADV process diagram of advertisement SLOT interface	21
Figure 8: Parameters for the step1 of the trigger function.	21
Figure 9: Parameters for the step2 of the trigger function.	22
Figure 10: Parameters for the step3 of the trigger function.	23
Figure 11: Description of the configured broadcasting process	23
Figure 12: Differences in the ADV process when the Lock Event Occurs ADV Duration feature is enabled or disabled.	24
Figure 13: Configuration of Environmental monitoring and analysis mode	25
Figure 14: Advertising workflow of Environmental monitoring and analysis mode	26
Figure 15: Advertising workflow of typical presence detection mode	27
Figure 16: Configuration of Movement-based dynamic tracking mode (APP)	28
Figure 17: Advertising workflow of typical dynamic tracking mode	29
Figure 18: Sensor configuration list	30
Figure 19: Configurations page of 3-axis accelerometer sensor	30
Figure 20: Configurations page of Temperature & Humidity sensor	31
Figure 21: Content of the exported Excel file	32
Figure 22: Configurations page of Quick switch	32
Figure 23: Hardware reset by button/magnet flowchart	34
Figure 24: How to hardware reset you Tag	35
Figure 25: Reset Battery warning page	36
Figure 26: Battery ADV mode selection page	36
Figure 27: ADV channel selection page	37
Figure 28: How to enable “Long Range” mode	37
Figure 29: Firmware upgrade workflow (Android)	39
Figure 30: Load upgrade package to nRF connect APP via iTunes	40
Figure 31: Firmware upgrade workflow of BeaconX Pro application (IOS)	40
Figure 32: Device information page	41
Figure 33: Obtain log file from Android APP	41

List of Tables

Table 1: Content structure of UID advertisement	10
Table 2: Content structure of URL advertisement	11
Table 3: Content structure of Customized – iBeacon advertisement	13
Table 4: Content structure of Customized – Sensor info advertisement	15
Table 5: Explanation of format parsing – Sensor info	16
Table 6: Different adverting periods in overall advertising logic	19
Table 7: Typical parameters configuration of Environmental monitoring and analysis mode	26
Table 8: Typical parameters configuration of presence detection mode	27
Table 9: Typical parameters configuration of movement-based dynamic tracking mode	29
Table 10: ADV type difference between 1M and Coded PHY	38

1. Overview

This User Guide is mainly applicable for MOKO standard **“BeaconX Pro”** APP and MK Sensor tag with **“BXP-S”** serial firmware, and mainly contained below parts:

- [Content parsing of Beacon advertisement](#)
- [“MK Sensor” APP user guidance](#)
- [FAQ](#)

For more hardware technical information or assembly/disassembly instructions, please ask our sales team directly for separated document – **“Product Specification”**.

2. Content parsing of Beacon advertisement

Before introducing the APP user manual, we will guide you to fully understand the Beacon advertisement format and content parsing rules. In terms of the MOKO “**BXP-S**” series firmware, which is especially design for asset tracking, it contains several kinds of advertisement frame type which include standard format and customized format, that is:

- *Google Eddystone™ - UID*
- *Google Eddystone™ - URL*
- *Google Eddystone™ - TLM (Unencrypted)*
- *Customized advertisement frame – “iBeacon”*
- *Customized advertisement frame - “Sensor info”*

To improve the readability of the screen formats, this document presents:

- The fixed Bluetooth fields highlighted in **blue**.
- The variable Bluetooth fields highlighted in **orange**.
- User-defined/configurable fields highlighted in **green**.
- Reserved fields highlighted in **red**.

The screenshots have been taken on the “**nRF Connect**” application, developed by Nordic Semiconductors.

2.1 Google Eddystone – UID

In terms of UID advertisement frame type, this is exactly complied with Google Eddystone regulations, and you can refer to below link for more information: <https://github.com/google/eddystone/tree/master/eddystone-uid>

Below table described the detailed content structure and frame example of **UID** advertisement:

Eddystone - UID																					
Frame bytes	Frame format		Frame example																		
	01	Length: 0x02	Received frame : Eddystone - UID <table><tr><td>Namespace ID</td><td>0x0102030405060708090A</td></tr><tr><td>Instance ID</td><td>0x102030405060</td></tr><tr><td>RSSI@0m</td><td>0x00</td></tr></table> Eddystone - UID field Raw data: 0x0201060303AAFE1716AAFE00000102030405060708090A102030405060000 Details: <table><tr><th>LEN.</th><th>TYPE</th><th>VALUE</th></tr><tr><td>2</td><td>0x01</td><td>0x06</td></tr><tr><td>3</td><td>0x03</td><td>0xAAFE</td></tr><tr><td>23</td><td>0x16</td><td>0xAAFE00000102030405060708090A1020304050600000</td></tr></table> LEN. - length of EIR packet (Type + Data) in bytes, TYPE - the data type as in https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile OK	Namespace ID	0x0102030405060708090A	Instance ID	0x102030405060	RSSI@0m	0x00	LEN.	TYPE	VALUE	2	0x01	0x06	3	0x03	0xAAFE	23	0x16	0xAAFE00000102030405060708090A1020304050600000
	Namespace ID	0x0102030405060708090A																			
	Instance ID	0x102030405060																			
	RSSI@0m	0x00																			
	LEN.	TYPE		VALUE																	
	2	0x01		0x06																	
	3	0x03		0xAAFE																	
	23	0x16		0xAAFE00000102030405060708090A1020304050600000																	
	02	Type: 0x01																			
	03	Data: 0x06																			
	04	Length: 0x03																			
	05	Type: 0x03																			
	06	Eddystone_UUID_LSB: 0xAA																			
	07	Eddystone_UUID_MSB: 0xFE																			
	08	Length: 0x17																			
	09	Type: 0x16																			
	10	Eddystone_UUID_LSB: 0xAA																			
	11	Eddystone_UUID_MSB: 0xFE																			
	12	Frame type UUID: 0x00																			
	13	RSSI@0m: 0x00																			
	14	Namespace ID[0]																			
	15	Namespace ID[1]																			
	16	Namespace ID[2]																			
	17	Namespace ID[3]																			
	18	Namespace ID[4]																			
	19	Namespace ID[5]																			
	20	Namespace ID[6]																			
	21	Namespace ID[7]																			
	22	Namespace ID[8]																			
	23	Namespace ID[9]																			
24	Instance ID[0]																				
25	Instance ID[1]																				
26	Instance ID[2]																				
27	Instance ID[3]																				
28	Instance ID[4]																				
29	Instance ID[5]																				
30	Reserved																				
31	Reserved																				

Table 1: Content structure of UID advertisement

2.2 Google Eddystone – URL

In terms of URL advertisement frame type, this is exactly complied with Google Eddystone regulations, and you can refer to below link for more information: <https://github.com/google/eddystone/tree/master/eddystone-url>

Below table described the detailed content structure and frame example of **URL** advertisement:

Frame bytes

Frame format	
01	Length: 0x02
02	Type: 0x01
03	Data: 0x06
04	Length: 0x03
05	Type: 0x03
06	Eddystone_UUID_LSB: 0xAA
07	Eddystone_UUID_MSB: 0xFE
08	Length: ≤ 0x17
09	Type: 0x16
10	Eddystone_UUID_LSB: 0xAA
11	Eddystone_UUID_MSB: 0xFE
12	Frame type UUID: 0x10
13	RSSI@0m: 0x00
14	URL Scheme Prefix: 0x00
15	Encoded URL[0]
16	Encoded URL[1]
17	Encoded URL[2]
18	Encoded URL[3]
19	Encoded URL[4]
20	Encoded URL[5]
21	Encoded URL[6]
22	Encoded URL[7]
23	Encoded URL[8]
24	Encoded URL[9]
25	Encoded URL[10]
26	Encoded URL[11]
27	Encoded URL[12]
28	Encoded URL[13]
29	Encoded URL[14]
30	Encoded URL[15]
31	Encoded URL[16]

Frame example

Received frame : Eddystone - URL

RSSI@0m	0x00
URL Scheme Prefix	0x00 = http://www.
Encoded URL	0x6D6F6B6F736D617274 = mokosmart (ASCII) 0x07 = .com (URL Expansion defined by Eddystone)

Eddystone - URL field

Raw data:

0x0201060303AAFE1016AAFE1000006D6F6B6F736D61727407

Details:

LEN.	TYPE	VALUE
2	0x01	0x06
3	0x03	0xAAFE
16	0x16	0xAAFE1000006D6F6B6F736D61727407

LEN. - length of EIR packet (Type + Data) in bytes,
TYPE - the data type as in <https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile>

OK

Table 2: Content structure of URL advertisement

2.3 Google Eddystone – TLM (Unencrypted)

In terms of TLM advertisement frame type, this is exactly complied with Google Eddystone regulations, and for more details you can refer to below link: <https://github.com/google/eddystone/blob/master/eddystone-tlm/tlm-plain.md>

Below table described the detailed content structure and frame example of **TLM** advertisement:

Eddystone - TLM																									
Frame bytes	Frame format		Frame example																						
	01	Length: 0x02	Received frame : Eddystone - TLM <table><tr><td>TLM version</td><td>0x00 = Unencrypted</td></tr><tr><td>Battery voltage</td><td>0x0C87 = 3207mV</td></tr><tr><td>Beacon temperature</td><td>0x0D80 = 13.5°C (8.8 fixed-point notation)</td></tr><tr><td>ADV_CNT (1 time/unit)</td><td>0x00000659 = 1625 times Remark: running count of advertisement frames of all types emitted by the beacon since power-up or reboot</td></tr><tr><td>SEC_CNT (0.1s/unit)</td><td>0x00003F6F = 1623.9s (00:27:03.900s) Remark: It is a 0.1 second resolution counter that represents time since beacon power-up or reboot</td></tr></table> Raw data: 0x0201060303AAFE1116AAFE20000C870D800000065900003F6F Details: <table><tr><th>LEN.</th><th>TYPE</th><th>VALUE</th></tr><tr><td>2</td><td>0x01</td><td>0x06</td></tr><tr><td>3</td><td>0x03</td><td>0xAAFE</td></tr><tr><td>17</td><td>0x16</td><td>0xAAFE20000C870D800000065900003F6F</td></tr></table> LEN. - length of EIR packet (Type + Data) in bytes, TYPE - the data type as in https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile OK Eddystone - TLM field	TLM version	0x00 = Unencrypted	Battery voltage	0x0C87 = 3207mV	Beacon temperature	0x0D80 = 13.5°C (8.8 fixed-point notation)	ADV_CNT (1 time/unit)	0x00000659 = 1625 times Remark: running count of advertisement frames of all types emitted by the beacon since power-up or reboot	SEC_CNT (0.1s/unit)	0x00003F6F = 1623.9s (00:27:03.900s) Remark: It is a 0.1 second resolution counter that represents time since beacon power-up or reboot	LEN.	TYPE	VALUE	2	0x01	0x06	3	0x03	0xAAFE	17	0x16	0xAAFE20000C870D800000065900003F6F
	TLM version	0x00 = Unencrypted																							
	Battery voltage	0x0C87 = 3207mV																							
	Beacon temperature	0x0D80 = 13.5°C (8.8 fixed-point notation)																							
	ADV_CNT (1 time/unit)	0x00000659 = 1625 times Remark: running count of advertisement frames of all types emitted by the beacon since power-up or reboot																							
	SEC_CNT (0.1s/unit)	0x00003F6F = 1623.9s (00:27:03.900s) Remark: It is a 0.1 second resolution counter that represents time since beacon power-up or reboot																							
	LEN.	TYPE		VALUE																					
	2	0x01		0x06																					
	3	0x03		0xAAFE																					
	17	0x16		0xAAFE20000C870D800000065900003F6F																					
	02	Type: 0x01																							
	03	Data: 0x06																							
	04	Length: 0x03																							
	05	Type: 0x03																							
	06	Eddystone_UUID_LSB: 0xAA																							
	07	Eddystone_UUID_MSB: 0xFE																							
	08	Length: 0x11																							
	09	Type: 0x16																							
	10	Eddystone_UUID_LSB: 0xAA																							
	11	Eddystone_UUID_MSB: 0xFE																							
	12	Frame type UUID: 0x20																							
	13	TLM version: 0x00																							
	14	Battery voltage (1mV)_LSB																							
	15	Battery voltage (1mV)_MSB																							
	16	Beacon temperature (0.1°C)_LSB																							
	17	Beacon temperature (0.1°C)_MSB																							
	18	ADV_CNT[0]																							
	19	ADV_CNT[1]																							
	20	ADV_CNT[2]																							
	21	ADV_CNT[3]																							
22	SEC_CNT[0]																								
23	SEC_CNT[1]																								
24	SEC_CNT[2]																								
25	SEC_CNT[3]																								
26	Not used																								
27	Not used																								
28	Not used																								
29	Not used																								
30	Not used																								
31	Not used																								

2.4 Customized frame – “iBeacon”

This *Customized iBeacon* advertisement frame is combined with two parts, of which one is standard APPLE iBeacon format which comply with APPLE iBeacon regulations, another one is a customized scan response packet.

What the purpose of adding extra scan response packet is just displaying more information on iOS APP, as APP cannot get the detail information from APPLE iBeacon advertisement due to iOS system restrictions. For APPLE iBeacon regulations, you can refer to the link: <https://developer.apple.com/ibeacon/Getting-Started-with-iBeacon.pdf>.

Below table described the detailed content structure and frame example of **Customized-iBeacon** advertisement:

MOKO Customized - iBeacon			
Frame bytes		Standard iBeacon packet	Scan response packet
	01	Length: 0x02	Length: 0x02
	02	Type: 0x01	Type: 0x0A
	03	Data: 0x06	Tx Power: 0x00
	04	Length: 0x1A	Length: 0x1A
	05	Type: 0xFF	Type: 0x16
	06	Apple CIN_LSB: 0x4C	Service data UUID_LSB: 0xAB (MOKO defined)
	07	Apple CIN_MSB: 0x00	Service data UUID_MSB: 0xFE (MOKO defined)
	08	Beacon type: 0x02	Frame type: 0x50 (MOKO defined)
	09	Data size: 0x15	RSSI@1m: 0xC5
	10	UUID[0]	ADV interval: 0x0A
	11	UUID[1]	UUID[0]
	12	UUID[2]	UUID[1]
	13	UUID[3]	UUID[2]
	14	UUID[4]	UUID[3]
	15	UUID[5]	UUID[4]
	16	UUID[6]	UUID[5]
	17	UUID[7]	UUID[6]
	18	UUID[8]	UUID[7]
	19	UUID[9]	UUID[8]
	20	UUID[10]	UUID[9]
	21	UUID[11]	UUID[10]
	22	UUID[12]	UUID[11]
	23	UUID[13]	UUID[12]
	24	UUID[14]	UUID[13]
	25	UUID[15]	UUID[14]
	26	Major[0]	UUID[15]
	27	Major[1]	Major[0]
	28	Minor[0]	Major[1]
	29	Minor[0]	Minor[0]
	30	RSSI@1m: 0xC5	Minor[0]
	31	Not used	Not used

Table 3: Content structure of Customized – iBeacon advertisement

Received frame : MOKO Customized - iBeacon	
UUID	0x0001020304050607080910A0B0C0D0E0F
Major	0x0001
Minor	0x0001
RSSI@1m	0xC5 = -59dBm
Tx Power	0x00 = 0dBm
ADV interval (100ms/unit)	0x0A = 1000ms

Raw data:

```
0x0201061AFF4C000215000102030405060708090A0B0C0D0E0F00010001C5020A001A16ABFE50C50A000102030405060708090A0B0C0D0E0F00010001
```

Details:

	LEN.	TYPE	VALUE
APPLE-iBeacon	2	0x01	0x06
	26	0xFF	0x4C000215000102030405060708090A0B0C0D0E0F00010001C5
Scan Response Packet	2	0x0A	0x00
	26	0x16	0xABFE50C50A000102030405060708090A0B0C0D0E0F00010001

LEN. - length of EIR packet (Type + Data) in bytes,
TYPE - the data type as in <https://www.bluetooth.org/en-us/specification/assigned-numbers/generic-access-profile>

OK

Standard iBeacon packet field

Tx Power

Scan response packet field

Figure 1: Example content structure of Customized – iBeacon advertisement

2.5 Customized frame – “Sensor info”

This “Sensor info” advertisement frame is customized by MOKO and also combined with two parts which is “advertising packet” and “scan response packet”.

- **ADV_IND (advertising packet)** ----Mainly to express the tag ID, sensor status, trigger event counts, battery voltage and acceleration data.
- **SCAN_RSP (scan response packet)** ----Mainly to express the device name.

Below table described the detailed content structure and frame example of **Customized-Sensor info** advertisement:

MOKO Customized - Sensor info			
Frame bytes		Advertisement packet	Scan response packet
	01	Length: 0x02	Length: ≤ 0x15
	02	Type: 0x01	Type: 0x09
	03	Data: 0x06	Device name[0]
	04	Length: 0x1B	Device name[1]
	05	Type: 0x16	Device name[2]
	06	Service data UUID_LSB: 0x01 (MOKO defined)	Device name[3]
	07	Service data UUID_MSB: 0xEA (MOKO defined)	Device name[4]
	08	Frame type UUID: 0x80 (MOKO defined)	Device name[5]
	09	Sensor status (Magnetic/ Motion/ Temperature/ Humidity/ Flash)	Device name[6]
	10	MAGNET_CNT_LSB	Device name[7]
	11	MAGNET_CNT_MSB	Device name[8]
	12	MOTION_CNT_LSB	Device name[9]
	13	MOTION_CNT_MSB	Device name[10]
	14	Accel.Dat_ X axis (mg) LSB	Device name[11]
	15	Accel.Dat_ X axis (mg) MSB	Device name[12]
	16	Accel.Dat_ Y axis (mg) LSB	Device name[13]
	17	Accel.Dat_ Y axis (mg) MSB	Device name[14]
	18	Accel.Dat_ Z axis (mg) LSB	Device name[15]
	19	Accel.Dat_ Z axis (mg) MSB	Device name[16]
	20	Temperature.Dat (°C) LSB	Device name[17]
	21	Temperature.Dat (°C) MSB	Device name[18]
	22	Humidity.Dat (%RH) LSB	Device name[19]
	23	Humidity.Dat (%RH) MSB	Not used
	24	Battery percentage(1%) / Battery voltage (1mV)_LSB	Not used
	25	Battery percentage(1%) / Battery voltage (1mV)_MSB	Not used
	26	Tag ID[0]	Not used
	27	Tag ID[1]	Not used
	28	Tag ID[2]	Not used
	29	Tag ID[3]	Not used
	30	Tag ID[4]	Not used
	31	Tag ID[5]	Not used

Table 4: Content structure of Customized – Sensor info advertisement

Received frame : MOKO Customized - Sensor info			
Sensor status	[Bit0]: Magnetic sensor working status	0011 1100'b (0x1C)	[Bit0] = 0 : Magnetic is approaching (RFU)
	[Bit1]: Motion sensor working status		[Bit1] = 0 : Device is in static
	[Bit2]: Motion sensor equipped status		[Bit2] = 1 : Motion sensor is equipped in the device
	[Bit3]: Temperature sensor equipped status		[Bit3] = 1 : Temperature sensor is equipped in the device
	[Bit4]: Humidity sensor equipped status		[Bit4] = 1 : Humidity sensor is equipped in the device
	[Bit5]: Flash equipped status		[Bit5] = 0 : Flash is not equipped in the device
	[Bit6-7]: Reserved		[Bit6-7]: Reserved
MAGNET_CNT ADV interval		0x0000 = 0 (Times of magnetic triggered) (RFU)	
MOTION_CNT		0x0002 = 2 (Times of motion triggered)	
Accel.Data_X axis		0x003A = 58mg	
Accel.Data_Y axis		0x0076 = 118mg	
Accel.Data_Z axis		0x03A7 = 935mg	
Temperature Data		Sampling temperature in 2 bytes (Signed integer), unit: 0.1℃/digit. Example content parse: 0x00 C8(HEX)>>>200(DEC)>>>20.0℃	
Humidity Data		Sampling humidity in 2 bytes (Unsigned integer), unit: 0.1%/digit. Example content parse: 0x01 37(HEX)>>>311(DEC)>>>31.1%RH	
Battery percentage(1%/unit) / Battery voltage (1mv/unit)		If value greater than 100, it represents the battery voltage. For example, 0x0CCC = 3276mV. If value less than or equal to 100, it represents the battery percentage. For example, 0x0064 = 100%.	
Tag ID		0x000001	
Device Name		0x4D4B2053656E736F72 = MK Sensor	

Raw data:

0x020106181601EA803C0000022CFF
ECFFF4041800DE015B0C100000010A
094D4B2053656E736F72

Details:

	LEN.	TYPE	VALUE
Sensor info ADV packet	2	0x01	0x06
	24	0x16	0x01EA803C0000022CFFECFFF4041800DE015B0C10000001
Scan response packet	10	0x09	0x4D4B2053656E736F72

LEN. - length of AD packet (Type + Data) in bytes,
TYPE - the data type as in [Assigned Numbers.pdf](#),
chapter 2.3: Common Data Types.

OK

Table 5: Explanation of format parsing – Sensor info

3. APP user guidance

Well, you have already familiar with the advertisement format and well to know how to parse the content in the advertisement, now we will guide you to know the *MK Sensor Series APP* and let's do some configuration on the Tag. **For the latest APP, please search "MK Sensor"(Android) and download from Google play store, search "BeaconX Pro"(IOS) and download from Apple store.**

3.1 Discover your tag

The smartphone with "*BeaconX Pro*" APP (Android or iOS) installed can be as a *Bluetooth Center* to scan and discover the advertisement packets of Bluetooth Low Energy (BLE) Beacons, and it will provide multiple strategies to filter the Bluetooth packets and that can help you to capture the required BLE peripherals directly and quickly.

3.1.1 Select the device type of your tag

Step 1: When enter into the app, the "*Beacon firmware options*" page will displayed, you can select the the "*MK-Sensor series*"option.

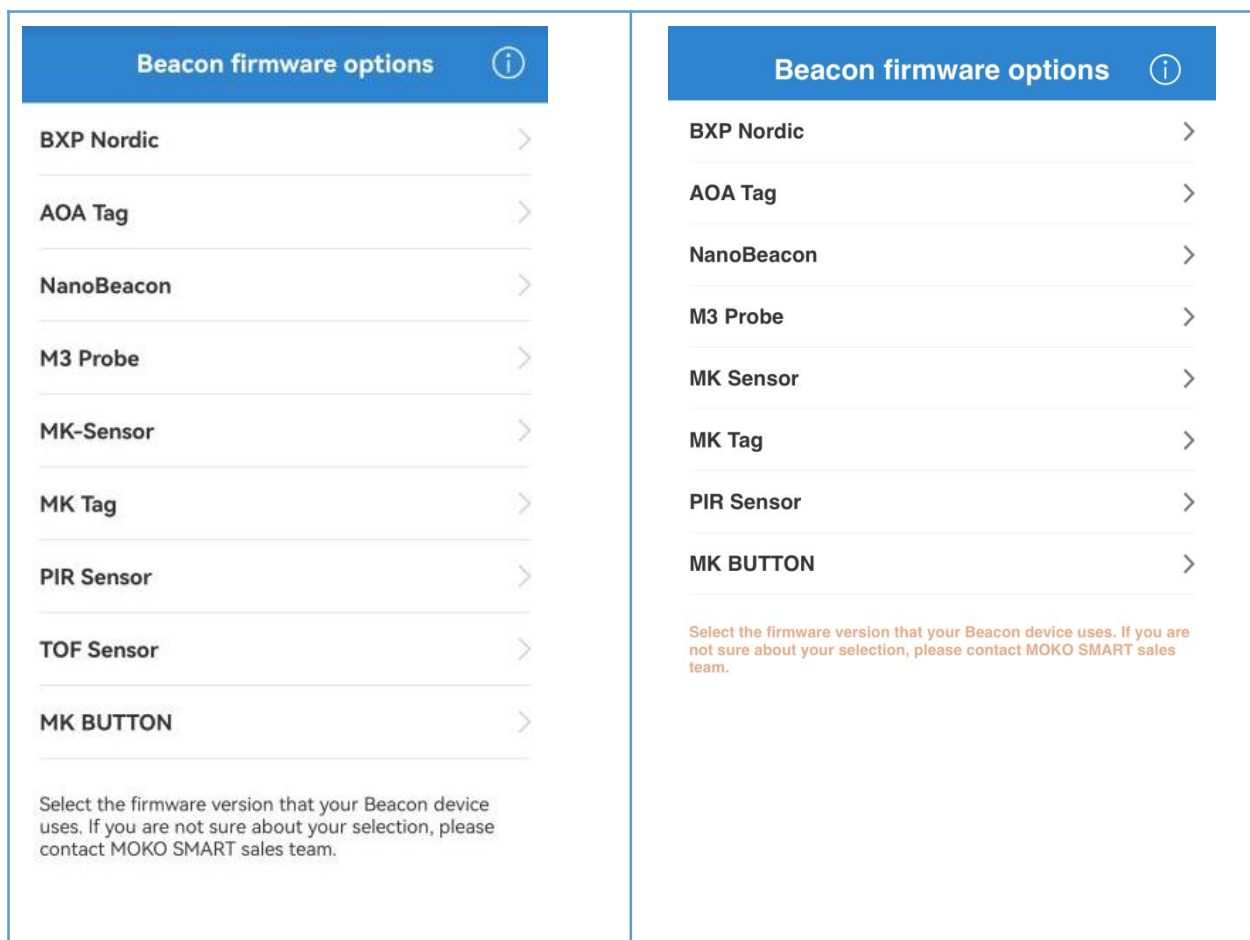


Figure 2: Beacon firmware options interface preview(Android&IOS)

3.1.2 Scanning your tag

After select the device type, the app page will enter into the “**SCANNING PREVIEW**” interface, the **SCANNER** will keep scanning continuously and list all surrounding advertising Bluetooth LE peripherals which match the device type, unless stop scanning manually by pressing scanning button “↺”.

Below picture shows the example scanning preview of MK Sensor series products.

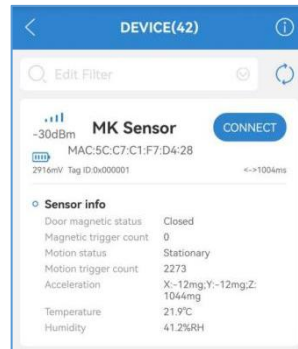
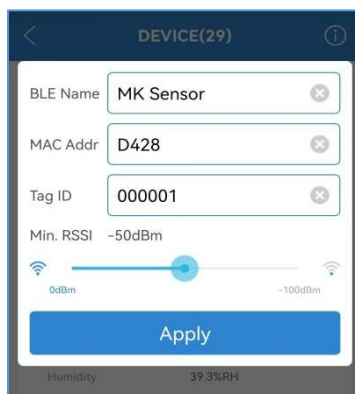


Figure 3: Example scanning preview of MK Sensor series

3.1.3 Filtering your tag

To find your tag quickly, we have developed multiple strategies to filter among mass Beacon devices, users may narrow your device list by BLE name, MAC address and RSSI value through “**Edit Filter**” option.



BLE Name: Allow to input 1-20 characters

Example: Only device's BLE name contain "MK Sensor" will be displayed in **SCANNER** interface.

NOTE:

1. Space effective, and **case insensitive** as well.
2. Only devices that have configured the Sensor Info frame and have the response packet enabled support broadcasting the device name.

MAC Addr: Allow to input 1-6 bytes (HEX)

Example: Only device's Mac address contain "B3:E8" will be displayed in **SCANNER** interface

NOTE: There is no need to input colon between MAC address in search filter column, and **case insensitive** as well

TAG ID: Allow to input 1-6 bytes (HEX)

Example: Only device's Tag ID contain "000001" will be displayed in **SCANNER** interface.

NOTE: Only devices that have configured the **Sensor Info** frame support broadcasting the Tag ID.

Min.RSSI: Only the advertisement packets with RSSI not less than the Min.RSSI value will be displayed in **SCANNER** interface, and configuration range is -100dBm to 0dBm

Example: When setting the Min.RSSI value to -70, the ADV packets with RSSI of -70dBm or greater than -70dBm will be list in **SCANNER** interface.

NOTICE:

1. **RSSI is the highest priority filtering condition. It means that filtered device must meet the RSSI filtering condition.**
2. **There is an "OR" relationship between BLE Name ,Mac Addr and Tag ID. It means that filtered device just need meet one of the filtering conditions.**
3. **The TAG ID is only included in the Sensor Info frame. Please confirm that the device you want to filter has enabled the corresponding type of broadcast frame.**



Figure 4: Configurations of scanning filter settings

3.2 Connect with your tag

For the Tag with “BXP-S” series firmware, it will display “**CONNECT**” button in the “**SCANNING PREVIEW**” interface if the status is connectable, when you Press “**CONNECT**” button, the tag’s LED will blink 2 times rapidly, then Input correct connection password and press “**OK**” button, You can successfully enter the device configuration interface.

Notice : Default password in MOKO standard firmware is “Moko4321”.

3.3 Know your tag’s advertising logic

Before setting up your tag’s advertisement, you may need to understand the basic advertising logic and working flowchart of the tag. With the integration of ultra-long advertising interval (standby duration) feature , in order to guarantee the efficiency of connections based on ultra-long advertising interval mode, we have integrated a constant advertising cycle which is called configuration period when device boot up.

The configuration period will last for 30 seconds and then device will start working (advertising period) by following the rules you preset in the advertisement SLOTS.

Overall advertising logic		
Advertising Period	Device status	Default Configurations
Configuration period	Connectable	Adv interval – 1000ms (non-configurable) Adv duration – 30s (non-configurable)
Advertising period	Connectable	Low-power mode – disable (configurable) Adv interval – 1000ms (configurable) Adv duration — 10s (configurable) Trigger function – disable (configurable)

Table 6: Different advertising periods in overall advertising logic

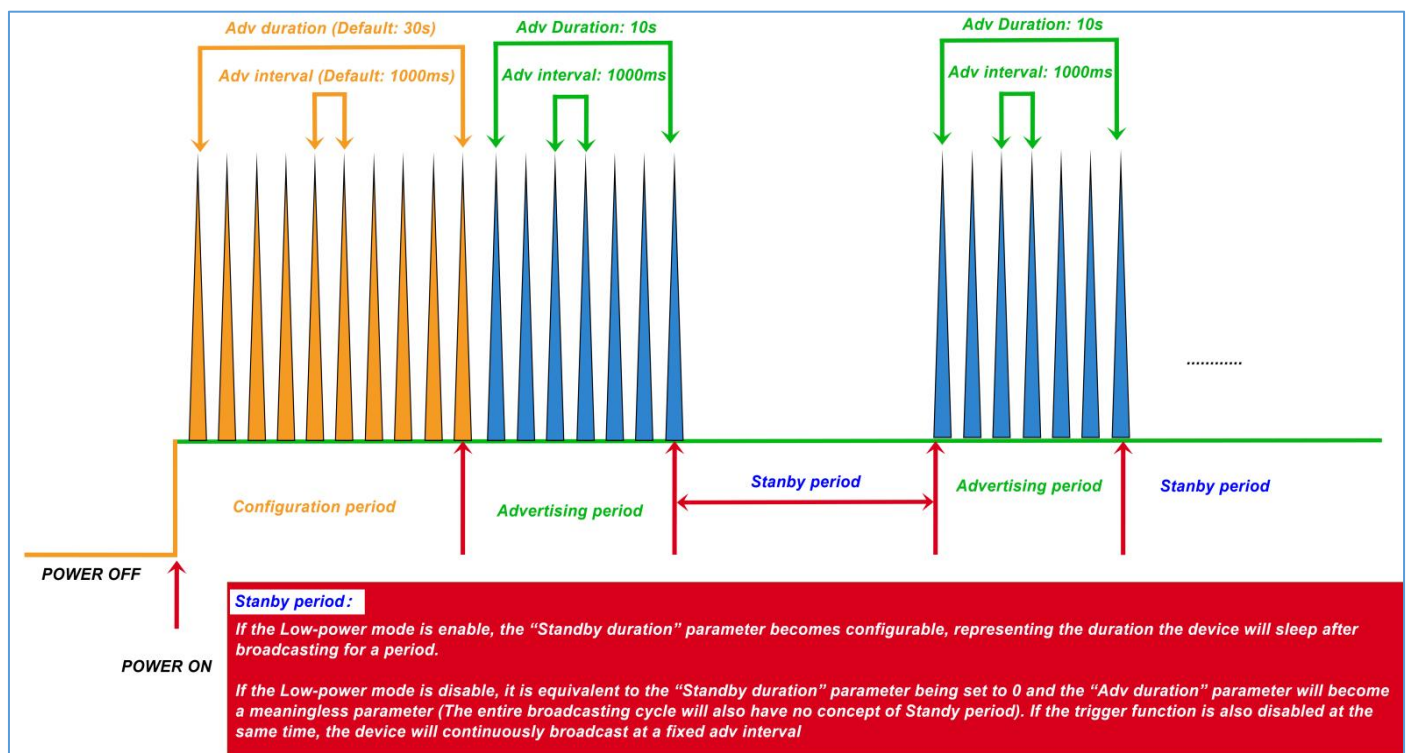


Figure 5: Overall advertising logic by default parameters

3.4 Access to SLOT configurations

Once device connected, the main screen of the APP will display the ***SLOT interface***. This interface is used to configure broadcast parameters of the device.

There have three advertisement slots can be configured separately in SLOT interface, and different device type will show different advertisement frame.

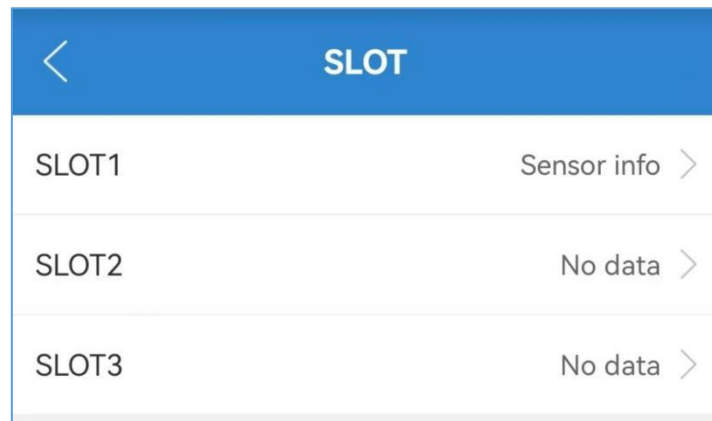


Figure 6: Configurations of advertisement SLOT interface

Remark: The tag can be configured with a maximum of three SLOTS. Six advertising types are supported. **No data** indicating that the SLOT is disabled.

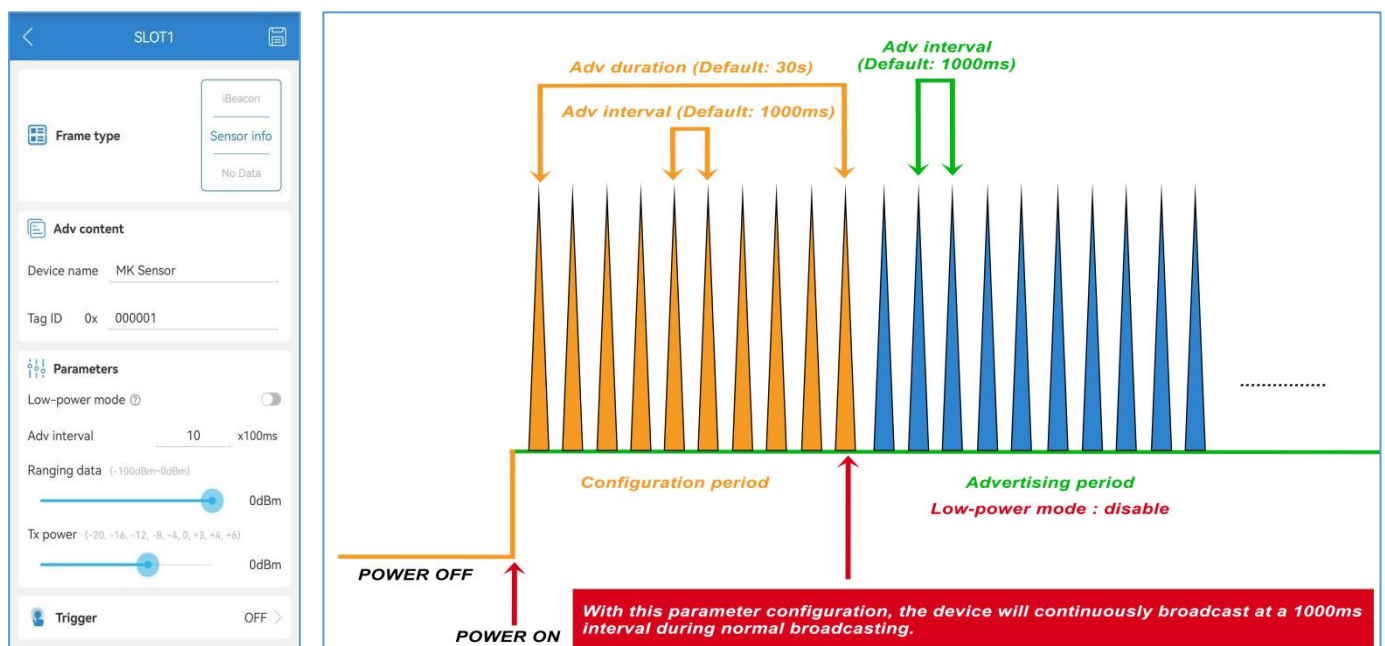


Notice – It is not recommended to configure all 3 slots as **No data**, that means the device will not broadcast any advertisement unless user reset the Beacon.

3.4.1 Setting your tag's advertisement SLOT

You can configure a certain frame type on different SLOT, each SLOT can be same frame type or different frame type.

Please refer to [Chapter 2 : Content parsing of Beacon advertisement](#) to know the parameters of the Adv content.



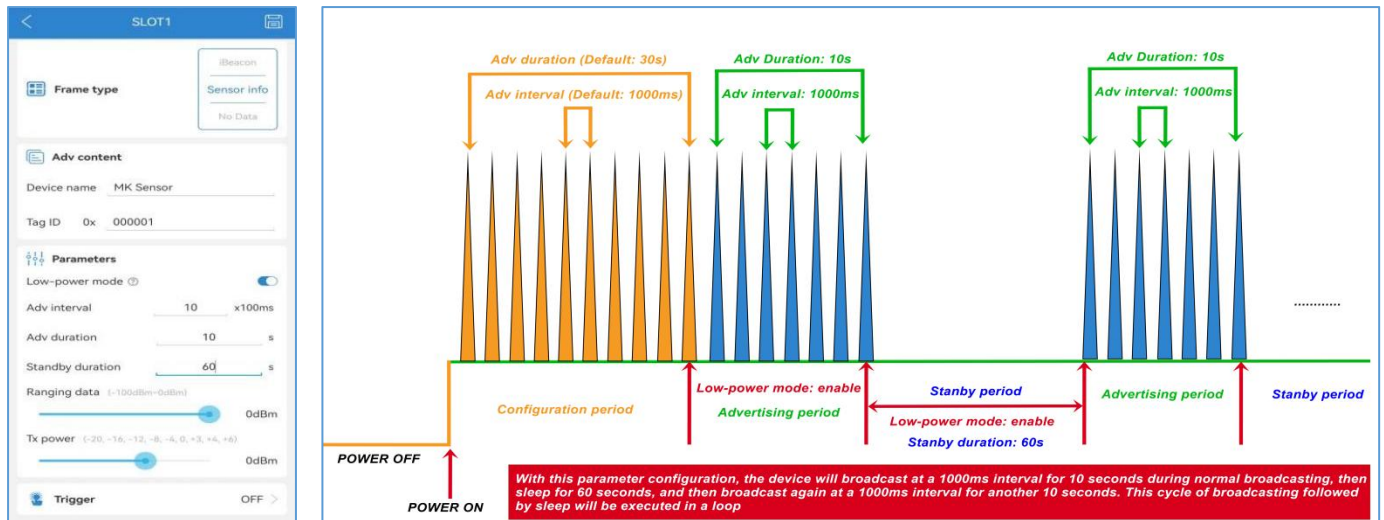


Figure 7: Parameters and ADV process diagram of advertisement SLOT interface

Low-power mode : The device follows a broadcast-sleep cycle operation mode. Initially, it broadcasts for a set period of time(**ADV Duration**) and then enters a low-power sleep state(**Standby period**). Upon receiving a trigger or reaching the scheduled wake-up time(**Standby duration**), it wakes up to broadcasts for a set period of time again. After completing the task, the device returns to sleep mode, minimizing energy consumption while remaining ready to operate when needed.

- If the Low-power mode is enable, the “**Standby duration**” parameter becomes configurable, representing the duration the device will sleep after broadcasting for a period.
- If the Low-power mode is disable (equivalent to “**Standby duration**” set to 0), and the “**Adv duration**” parameter will become a invalid parameter (The entire broadcasting cycle will also have no concept of Standby period).

3.4.2 Setting your tag's trigger function

Trigger function is designed for some emergency states switching or some specific defined use cases, which include **Motion trigger**, **Temperature trigger** and **Humidity trigger**. You can identify the different custom defined event through different advertising status before and after triggered.

Click the “**Trigger**” option, then it will automatically jump to the trigger function configuration guide interface.

Step 1: Initial Setting. In this step, you can configure parameters for tags to broadcast after the trigger event occurs. The “**Trigger**” button must be enabled to processed to the next step.

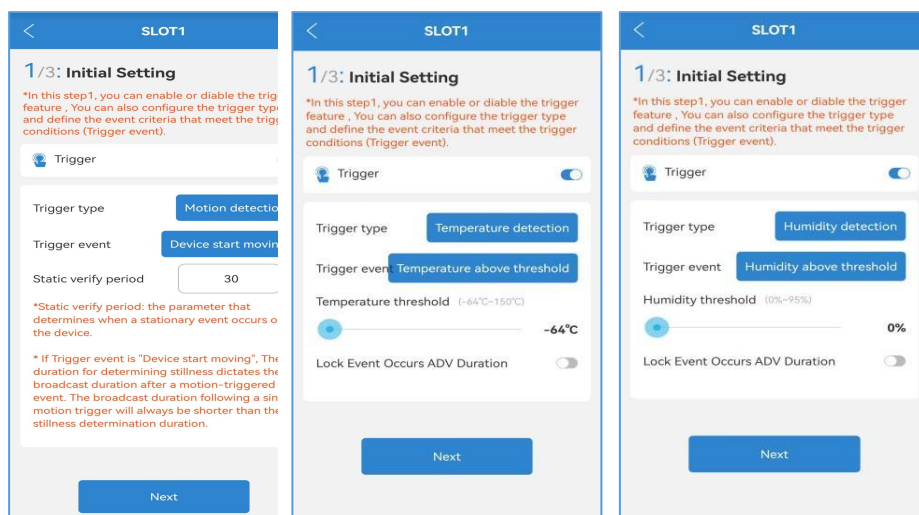


Figure 8: Parameters for the step1 of the trigger function.

- **Trigger button:** This button must be turned on, otherwise you will not be able to proceed to the next step.
- **Trigger type:** Motion detection / Temperature detection / Humidity detection / Magnetic detection
- **Trigger event:** Indicates the conditions for determining the occurrence of an event. If the event occurs, broadcasting will be carried out according to the broadcasting parameters specified in **Step 2**.

Note:

- **Static verify period:** The stationary detection duration is only applicable to the **"Motion detection"** type and indicates the condition for determining the occurrence of a stationary event. If Trigger event is **"Device start moving"**, The duration for determining stillness dictates the broadcast duration after a motion-triggered event. The broadcast duration following a single motion trigger will always be shorter than the stillness determination duration.
- **Lock Event Occurs ADV Duration:** This function is only applicable to the **"Temperature detection / Humidity detection / Magnetic detection"**. If the device quickly returns to a state where the triggering condition is no longer met after initially satisfying the triggering condition, it can only broadcast for a short duration, or might not broadcast at all. The Locked ADV function ensures that, in such cases, the set post-trigger broadcast duration is fully executed, regardless of changes in the triggering condition. If the **"Total adv duration"** in **step2** is set to 0, the Lock Event Occurs adv duration will default to a locked broadcast of 5 seconds. (Please refer to the [Chapter 3.4.3 Lock Event Occurs ADV Duration](#) for a more detailed understanding of this feature)

Step 2: Event occurs setting. In this step, you can configure the advertising parameters of trigger event occurs. The broadcasting parameters configured in this step are related to the **"Trigger event"** parameters configured in **Step 1**. For example, if you configure the **"Trigger event"** as **'Device starts moving'**, the device will broadcast according to the parameters configured in this step after it starts moving.

Frame type: The broadcast frame type after the trigger event occurs

Configurable item: Sensor info/ iBeacon/ Eddystone-TLM/ Eddystone-UID/ Eddystone-URL

Adv content: The broadcast content after the trigger event occurs. The configurable broadcast content varies depending on the frame type of broadcast

Adv interval: The broadcast interval after the trigger event occurs

Total adv duration: The broadcast duration after the trigger event occurs.

NOTE:

1. If the value is set to 0, It means that the device will continue broadcasting after the trigger event occurs, until the trigger state changes.
2. If Trigger event is **"Device start moving"**, The duration for determining stillness dictates the broadcast duration after a motion-triggered event. The **'Total adv duration'** configured for a single motion trigger is always shorter than the stationary detection duration.

Figure 9: Parameters for the step2 of the trigger function.

Step 3: Before event occurs setting. If you want the device to broadcast before the trigger event occurs, you can enable this feature. In this step, you can set the broadcasting parameters related to the time before the trigger event happens. The state before the trigger event refers to the opposite state of the trigger event you configured in Step 1. For example, if the trigger event is configured as 'device starts moving,' the state before the trigger event occurs refers to the device being stationary; if the trigger event is configured as 'door open,' the state before the trigger event occurs refers to the door being closed.

Advertising before trigger event occurs: If you want the device to broadcast before the trigger event occurs, you can enable this feature.

Frame type: The broadcasting frame type before the trigger is by default the same as the broadcasting frame type in Step 2 and cannot be configured, only the relevant broadcasting parameters can be configured.

Adv content: The broadcast content after the trigger event occurs. The configurable broadcast content varies depending on the frame type of broadcast

Low-power mode: This mode supports the device broadcasting in a loop (broadcast -> sleep -> broadcast -> sleep) while in the state before the trigger event occurs (the opposite state of the trigger event).

If the Low-power mode is enable, the "Standby duration" parameter becomes configurable, representing the duration the device will sleep after broadcasting for a period.

If the Low-power mode is disable (equivalent to the value of "Standby duration" being 0), and the "Adv duration" parameter will become a meaningless parameter (The entire broadcasting cycle will also have no concept of Standby period).

Adv interval: The broadcast interval after the trigger event occurs

Adv duration: Refers to the **broadcast time** of the device during the loop of broadcast -> sleep -> broadcast -> sleep.

Standby duration: Refers to the **sleep time** of the device during the loop of broadcast -> sleep -> broadcast -> sleep.

Figure 10: Parameters for the step3 of the trigger function.

Once all configuration steps are completed, click the **"Done"** button. A reminder box will pop up, which mainly provides an explanation of the device's broadcasting process. If you confirm there are no issues, click **'OK'** and the app will send the relevant parameters to configure the device accordingly.

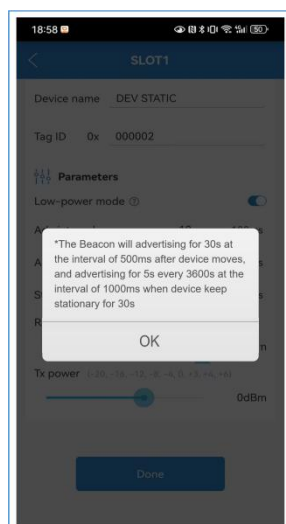


Figure 11: Description of the configured broadcasting process

You can also refer to [Chapter 3.5 How to apply to your scenarios](#) to learn more about detailed application scenarios, related configuration parameters and related broadcasting flowchart.

3.4.3 Lock Event Occurs ADV Duration

*The “**Lock Event Occurs ADV Duration**” function is only applicable to the “**Temperature detection / Humidity detection / Magnetic detection**”. If the device quickly returns to a state where the triggering condition is no longer met after initially satisfying the triggering condition, it can only broadcast for a short duration, or might not broadcast at all. The Locked ADV function ensures that, in such cases, the set post-trigger broadcast duration is fully executed, regardless of changes in the triggering condition.*

*If the “Total adv duration” in **step2** is set to 0, the “Lock Event Occurs ADV Duration” will default to a locked broadcast of 5 seconds. (Please refer to the below to understand the differences in whether this function is enabled)*

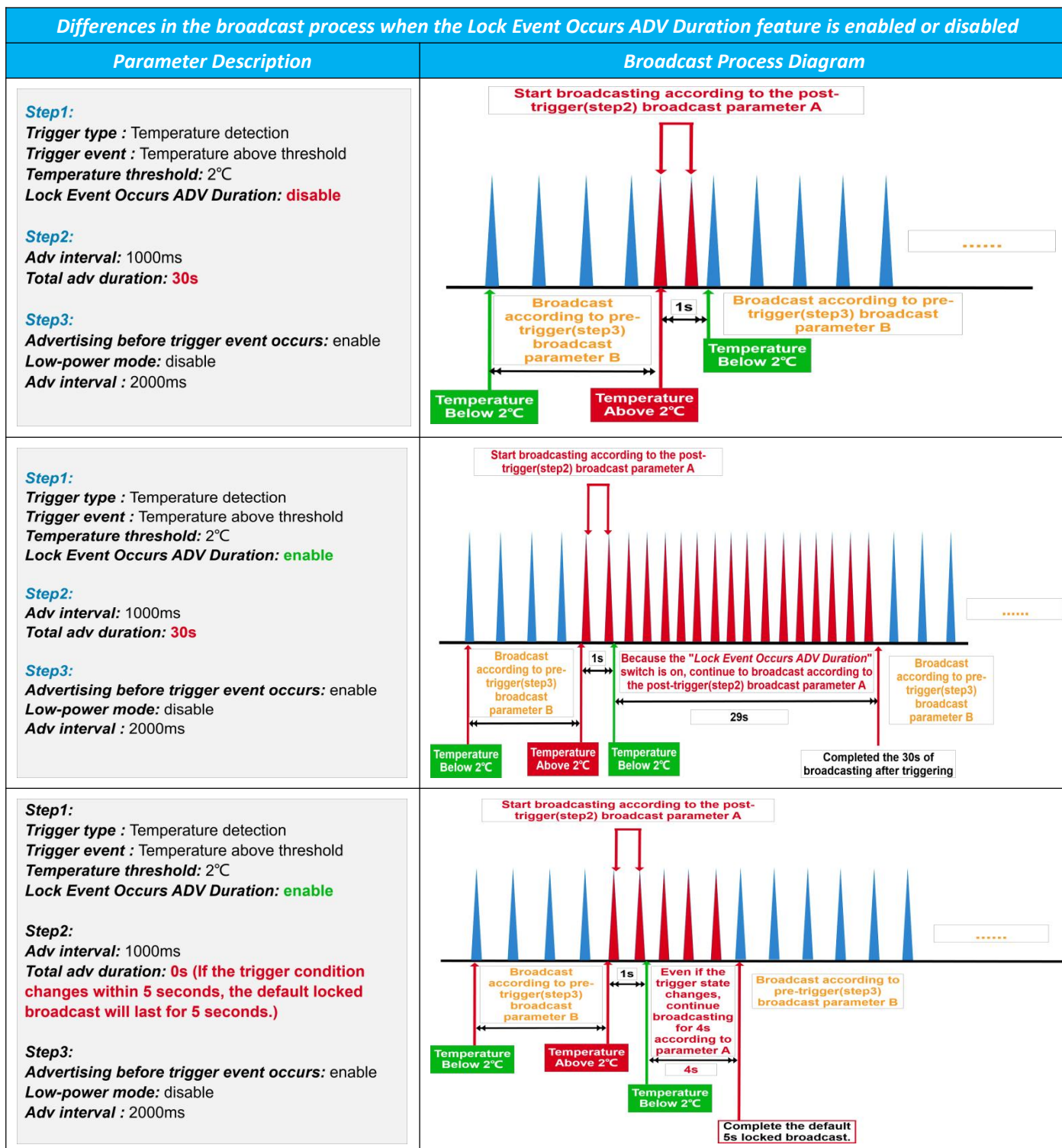


Figure 12: Differences in the ADV process when the Lock Event Occurs ADV Duration feature is enabled or disabled

3.5 How to apply to your scenarios

To expand the usage scenarios of “MK Sensor” series products in different scenarios, we have implemented some specific features to make it preferentially and exclusively adapted to various application scenarios.

- Environmental monitoring and analysis
- Asset management
 - Presence detection mode
 - Movement-based dynamic tracking mode

Here we will introduce some typical scenarios for your reference based on practical cooperation with end users. ***In order to ensure the feasibility of settings for all users in different application scenarios, we do not directly define the mode switch but need users to configure relevant parameters to achieve this.***

3.5.1 Environmental monitoring and analysis

This environmental monitoring mode is commonly useful for application such as cold-chain transport vehicle/ cold storage in cold-chain logistics, vaccine transport storage temperature monitoring in medical industry. fire alarm in factory or office.

In these practical scenarios, you must pay more attention to the periodic temperature data, especially the alarm information above the temperature threshold, and in order to save battery consumption, you can lower the advertisement report frequency when the temperature below the temperature threshold. In addition, the device may also need to periodically store temperature data for later analysis.

Below described the typical configurations and working logical of environmental monitoring and analysis mode:

Figure 13: Configuration of Environmental monitoring and analysis mode

Storage interval: The value set to 0 means that the storage interval follows the “Sampling interval”, and the temperature/humidity data will be stored every time a sample is taken.

Typical scenario 1 – Environmental monitoring and analysis mode			
items	Descriptions		Configurations
Working logical	Device will broadcast rapidly when the temperature above the temperature threshold and broadcast slowly when the temperature below the temperature threshold. The device will store temperature data to the flash every 15 minutes		-
Typical configurations SLOT 1	Trigger function		enable
	Step1	Trigger type	Temperature detection
		Trigger mode	Temperature above threshold
		Temperature threshold	2 °C
		Lock Event Occurs ADV Duration	enable
	Step2 (ParameterA)	Frame type	Sensor info
		Device name	TEMP ABOVE
		Tag ID	0001
		Advertising interval	1000ms
		Total adv duration	0s
	Step3 (ParameterB)	Advertising before triggered	enable
		Frame type	Sensor info
		Device name	TEMP BELOW
		Tag ID	0002
		Low-power mode	enable
		Advertising interval	1000ms
		Advertising duration	5s
		Standby duration	300s
		Tx Power	0dBm
Temperature sensor	Necessary		-
Data Storage	Temperature Data Store		enable
	Sampling interval		5min
	Storage interval		15min
Suggested scenarios	Cold-chain logistics –cold chain transport vehicle/ cold storage Medical industry – vaccine transport storage temperature monitoring		-

Table 7: Typical parameters configuration of Environmental monitoring and analysis mode

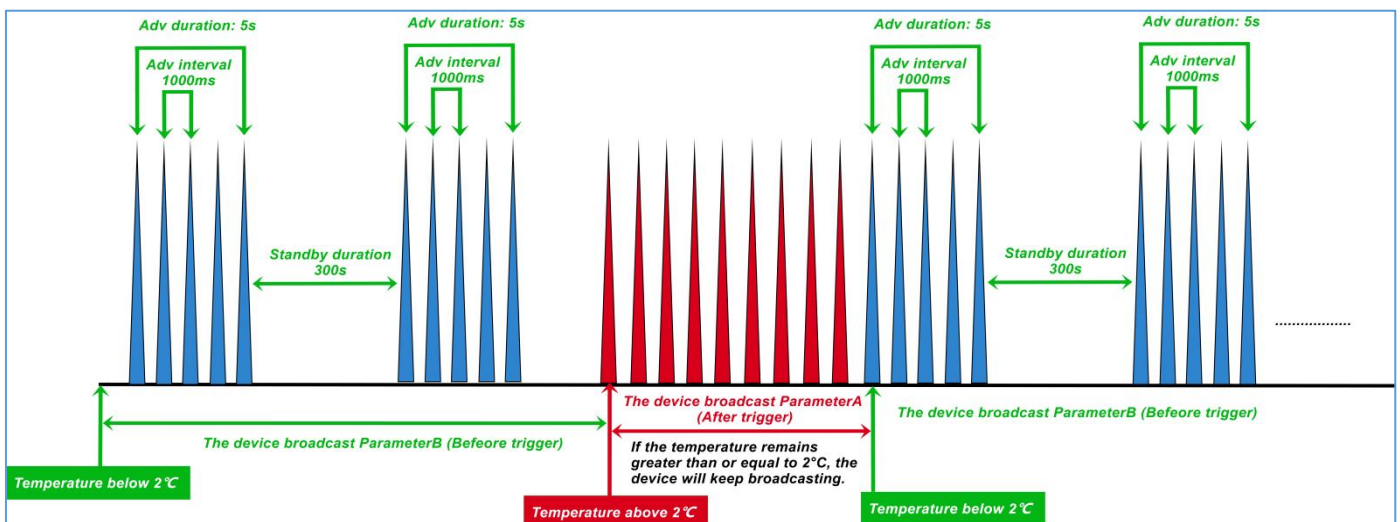


Figure 14: Advertising workflow of Environmental monitoring and analysis mode

3.5.2 Asset management

➤ Presence detection mode

This presence detection mode is commonly used for fixed assets, such as furniture/desks/photocopiers in the office, or medical equipments/beds in the hospital.

In these practical scenarios, you will only need to know it is in right place rather than the real-time positions, so many end users prefer to keep the device broadcasting every 1 hour or more.

Below described the typical configurations and working logical of presence detection mode:

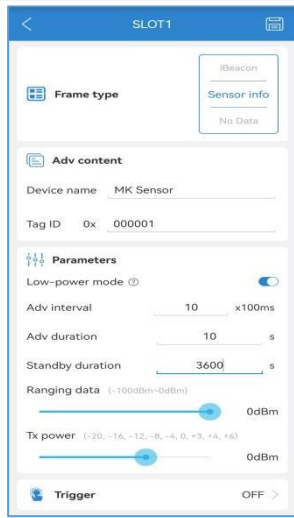
Typical scenario 2 – Presence detection mode			
items	Descriptions	Configurations	
Working logical	Device will broadcast 10s with 1000ms advertising interval every 1 hour		
Typical configurations SLOT 1	Frame type	Sensor info	
	Device name	MK Sensor	
	Tag ID	0001	
	Low-Power mode	enable	
	Advertising interval	1000ms	
	Advertising duration	10s	
	Standby duration	3600s	
Trigger function	Disable		
Motion sensor	Not necessary	-	
Suggested scenarios	Smart office – Furniture/desks/photocopiers tracking Healthcare – Medical equipment/beds tracking		

Table 8: Typical parameters configuration of presence detection mode

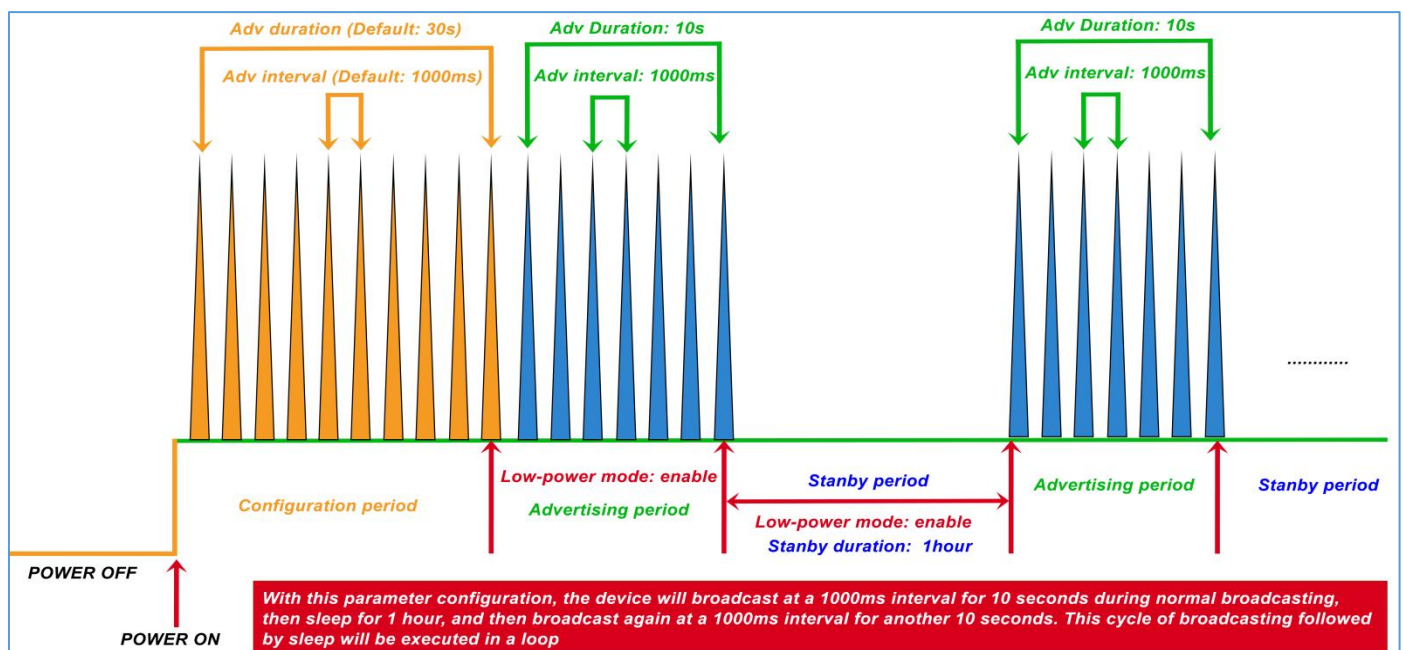


Figure 15: Advertising workflow of typical presence detection mode

➤ Movement-based dynamic tracking mode

This dynamic tracking mode is commonly used for RTLS application scenarios, such as packaging box/container/pallet in the warehouse, frequently used IV infusion pumps/wheelchairs in the hospital, or shopping carts in the market.

In these practical scenarios, you must need to know its real-time positions rather than simple presence status, and in order to save the battery consumption when assets in no use status (detected by motion sensor), you can lower the advertisement report frequency when assets in long-terms idle status. Based on these demands, many clients prefer the movement-based dynamic tracking mode.

Below described the typical configurations and working logical of dynamic tracking mode:

The figure displays three sequential screenshots of the MOKO SMART APP interface for configuring the Movement-based dynamic tracking mode. Each screen is titled 'SLOT1' and shows a different step in the configuration process.

1/3: Initial Setting
 *In this step1, you can enable or dable the trigger feature , You can also configure the trigger type and define the event criteria that meet the trigger conditions (Trigger event).
 Trigger: ☒
 Trigger type: Motion detection
 Trigger event: Device start moving
 Static verify period: 30 s
 *Static verify period: the parameter that determines when a stationary event occurs on the device.
 * If Trigger event is "Device start moving", The duration for determining stillness dictates the broadcast duration after a motion-triggered event. The broadcast duration following a single motion trigger will always be shorter than the stillness determination duration.

2/3: Event occurs setting
 *In this step, you can configure the advertising parameters of trigger event occurs.
 Frame type: Sensor info
 Adv content: Device name: Moving, Tag ID: 0x 000001
 Parameters: Adv interval: 5 x100ms, Total adv duration: 0 s, Ranging data: (-100dBm~0dBm), Tx power: (-20, -16, -12, -8, -4, 0, +3, +4, +6) 0dBm

3/3: Before event occurs setting
 *In this step, you can configure whether to enable pre-trigger advertising and related advertising parameters
 Advertising before trigger event occurs: ☒
 Frame type: Sensor info
 Adv content: Device name: Static, Tag ID: 0x 000002
 Parameters: Low-power mode: ☒, Adv interval: 10 x100ms, Adv duration: 5 s, Standby duration: 3600 s, Ranging data: (-100dBm~0dBm), Tx power: (-20, -16, -12, -8, -4, 0, +3, +4, +6) 0dBm

Figure 16: Configuration of Movement-based dynamic tracking mode (APP)

Table 9: Typical parameters configuration of movement-based dynamic tracking mode

3.6 Access to sensor configurations

The sensor configuration options are available on the “**SETTING >>>Sensor configurations**” page. On this page, you will find a list of all the sensors that support configuration.

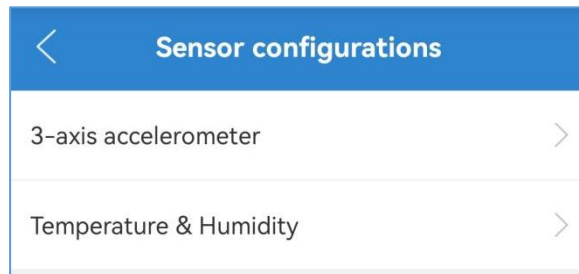


Figure 18: Sensor configuration list

3.6.1 Motion sensor (3-axis accelerometer)

Regarding of MK Sensor series product, currently we have adopted motion sensor for “*Movement-based dynamic tracking*” functions which link to “[Chapter 3.4.2 : Movement-based dynamic tracking mode](#)” for your reference, and you can access to “**SETTING >>>Sensor configurations>>> 3-axis accelerometer**” interface for further configurations.

Remark: Configuration interface will be shown only when device equipped with 3-axis accelerometer sensor.

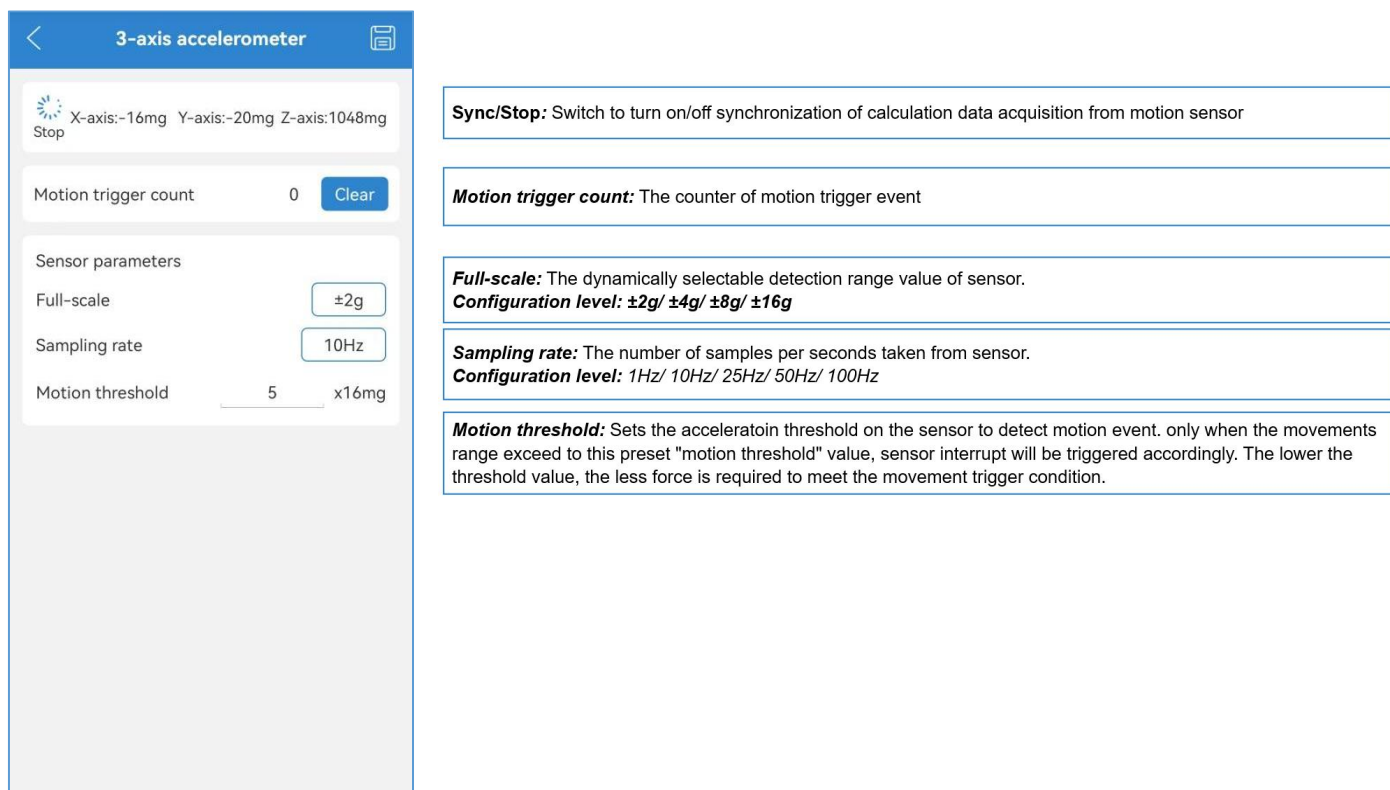


Figure 19: Configurations page of 3-axis accelerometer sensor

3.6.2 Temperature & Humidity sensor

Regarding of MK Sensor series product, currently we have adopted temperature and humidity sensor for “Environment monitoring and analysis” functions which link to [“Chapter 3.4.1 : Environmental monitoring and analysis”](#) for your reference, and you can access to **“SETTING >>>Sensor configurations>>> Temperature & Humidity”** interface or **“SETTING >>>Sensor configurations>>> Temperature”** interface for further configurations.

Remark:

1. **“SETTING >>>Sensor configurations>>> Temperature & Humidity”** configuration interface will be shown only when device equipped with temperature and humidity sensor .
2. **“SETTING >>>Sensor configurations>>> Temperature”** configuration interface will be shown only when device equipped with single temperature sensor .

Temperature & Humidity

Temperature
21.1 °C

Humidity
34.0 %RH

Sampling interval
5 sec (1-65535)

T&H Data Store
☒

Storage interval
5 min (0-65535)

Sync Beacon time
02/01/1970 03:27:12
Sync

Export T&H data

Temperature: Real-time temperature data

Humidity: Real-time humidity data

Sampling interval: The time in seconds when the sensor takes a temperature and humidity measurement

T&H Data Store: Temperature and humidity data storage switch. After this function is enabled, the device will store temperature and humidity data at the specified **Storage interval**

Storage interval: The interval time in minutes to store the temperature and humidity sampling data. For example, if you set it as 1min, then device will store the data every 1 minutes. The **Storage interval** can be set only if the **T&H Data Store** function is enabled.

Notice: When the storage interval is set to 0, it means the storage interval follows the sampling interval, storing data once for each sampling.

Sync Beacon time: Sync Beacon the smartphone time, and then system updated time will be displayed

Export T&H data: Click to enter this option, and you can access the stored historical temperature and humidity data, as well as perform other operations, for example

1. View all historical temperature and humidity data
2. Supports viewing temperature and humidity data in a graphical curve format.
3. Filter all the retrieved historical data by time period (you need to first retrieve all the data before you can filter by time period).
4. Erase all historical data
5. Export the historical temperature and humidity data as an Excel document, which includes all/filter-selected historical data as well as summary and analysis data.

Export T&H data

Sync
Display
Erase all
Export

Start Date: 2024-12-18 16:40:27
End Date: 2024-12-18 16:50:06
Sum records:49
Filtered records:10

Time	Temperature(°C)	Humidity(%RH)
18/12/2024 16:49:45	20.3	36.1
18/12/2024 16:48:45	20.3	36.2
18/12/2024 16:47:45	20.4	36.2
18/12/2024 16:46:45	20.4	36.2
18/12/2024 16:45:45	20.4	36.2
18/12/2024 16:44:45	20.5	36.3
18/12/2024 16:43:45	20.5	36.3
18/12/2024 16:42:45	20.6	36.3
18/12/2024 16:41:45	20.6	36.3
18/12/2024 16:40:45	20.6	36.3

Sync/Stop: Start Sync/Stop sync to retrieve all historical temperature and humidity data

Display: Viewing temperature and humidity data in a graphical curve format

Erase all: To erase all stored historical temperature and humidity data

Export: Export the historical temperature and humidity data as an Excel document, which includes all/filter-selected historical data as well as summary and analysis data.

Start Date&End data: Filter by time period to display or export the synced historical temperature and humidity data

Figure 20: Configurations page of Temperature & Humidity sensor

The data content in the exported Excel file is shown in the image below

Time	°C/°F	%RH	MAC Address	5CC7C1F7D428
18/12/2024 16:49:45	20.3 / 68.5	36.1	Sampling interval	1 s
18/12/2024 16:48:45	20.3 / 68.5	36.2	Storage interval	1 min
18/12/2024 16:47:45	20.4 / 68.7	36.2	Start Time	18/12/2024 16:40:45
18/12/2024 16:46:45	20.4 / 68.7	36.2	End Time	18/12/2024 16:49:45
18/12/2024 16:45:45	20.4 / 68.7	36.2	Total Duration Recorded	9Mins0Sec
18/12/2024 16:44:45	20.5 / 68.9	36.3	Total Data Recorded	10
18/12/2024 16:43:45	20.5 / 68.9	36.3	Max Temperature	20.6 / 69.1
18/12/2024 16:42:45	20.6 / 69.1	36.3	Min Temperature	18/12/2024 16:40:45
18/12/2024 16:41:45	20.6 / 69.1	36.3	Average Temperature	20.3 / 68.5
18/12/2024 16:40:45	20.6 / 69.1	36.3	Max Humidity	18/12/2024 16:40:45
			Min Humidity	36.1
			Average Humidity	18/12/2024 16:49:45
				36.2

Figure 21: Content of the exported Excel file

3.7 Quick switch for basic features

In current firmware, we have designed multiple function switch for some basic functions. Based on that, you can disable/enable these features through the switch, thus adapting to different use cases.

You can enable/disable these function switch in “**SETTING >>> Quick switch**” section, and below shows the “**Quick switch**” interface as well as relevant feature descriptions.

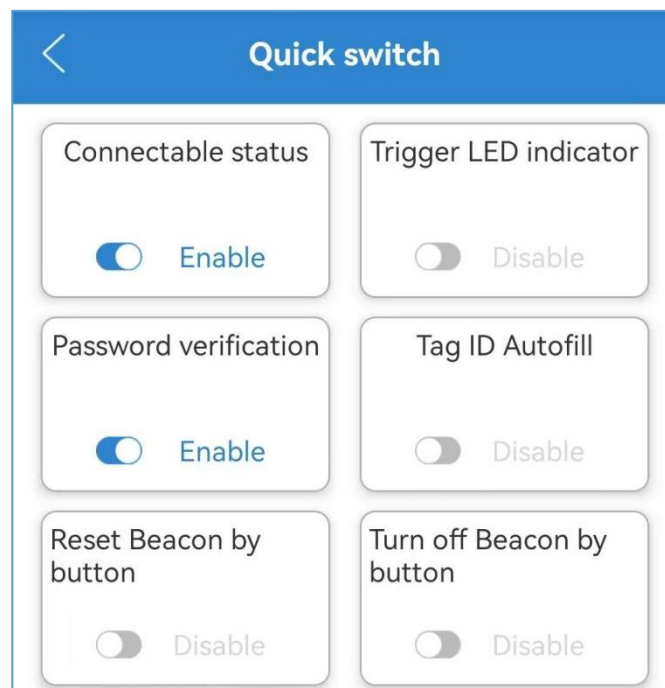


Figure 22: Configurations page of Quick switch

3.7.1 Switch your Tag to non-connectable status

In some cases, you will need to disable the tag's connectable property, to let it work just as a location tag. So you can disable the connectable status in "**SETTING >>> Quick switch >>> Connectable status**". After feature disabled, device won't be connected by master BLE device anymore unless reset your tag.

3.7.2 Disable trigger LED indicator

If you preset the motion trigger or magnetic trigger mechanism, the LED will blink for 1 time when it is triggered. Considering the invisibility in some specific scenarios or the save of battery consumption, you can access to "**SETTING >> Quick switch >> Trigger LED indicator**" to disable the LED notification in trigger mode.

3.7.3 Disable password verification

You can go to "**DEVICE >> Quick switch >> Password verification**" to disable password verification. Once password verification be disabled, there will be no need to input password when you connect with your tag.

Remark: If password verification feature was disabled, you will not be able to change password or reset Beacon due to security process restrictions.

3.7.4 Autofill Tag ID

As we have described previously, MOKO customized advertisement **Sensor info** format contained Tag ID. If you want the Tag ID and mac address to be the same, you can enable this feature and The Tag ID field of the **Sensor info** is automatically filled with beacon mac address.

You can access to "**DEVICE >> Quick switch >> Tag ID Autofill**" to enable it and then relevant information which included in scan response packet will not be shown in the "**SCANNING PREVIEW**" interface.

3.7.5 Disable/Enable your Beacon's button function

Basically the independent button in the Beacon was designed to realize power off or reset functions, but in some specific cases, there will need to disable the button functions in order to avoid intended shutdown or reset operations. So logically there also have the switch to enable/disable these functions.

You can access to **SETTING >> Quick switch >> Turn off Beacon by button** and **SETTING >> Quick switch >> Reset Beacon by button** to disable/enable these two button control functions.

3.8 Factory reset your Beacon

3.8.1 Remote reset

In some cases that mechanical button function should be disabled, to avoid intentional reset situations. Under this circumstance we will need remote reset function for backup. You can remote reset your tag by just connect with your tag and tapping "**SETTING >>> Reset Beacon**" option, then it will give prompt to you to confirm the reset operations.

Remark: Remote reset operations will not change the connection password to factory reset status. If you want to totally reset the Beacon with password restore as well, then you may need to do hardware reset operations.

3.8.2 Hardware reset

If your device is already in a non-connectable state, you will not be able to perform the remote reset operation mentioned in the previous section. Then you can try to perform a factory reset on your device using a hardware reset. Before performing a hardware reset, please make sure your device is loaded with the appropriate hardware.

➤ Hardware reset by button

If your device has a external or inner button and the “**SETTING >> Quick switch >> Reset Beacon by button**” option is turned off, you can try using the button to perform a factory reset. Below is the operation flow:

Step1: Keep the device power off. If the device is turned on, please press and hold the button for 3 seconds to turn it off.

Step2: Press and hold the button for 10 seconds or more, during this operation the device will power on and LED start flashing for 3 seconds.

Step3: After long press duration more than 10 seconds, release the button and then press the button again within 2 seconds.

Step4: The LED on the device will keep solid red for 3 seconds, and then the device will automatically restart. After completing the above steps, you are done restoring factory Settings.

➤ Hardware reset by magnet:

If your device has an inner hall sensor switch and the “**SETTING >> Quick switch >> Reset Beacon by button**” option is turned off, you can try using the magnet to perform a factory reset. Below is the operation flow:

Step1: Keep the device power off. If the device is turned on, hold a magnet close to the position marked with a Hall switch sign on the product and keep it there for 3 seconds to turn it off, then remove the magnet and keep it away from the device .

Step2: hold the magnet close to the position marked with a Hall switch sign on the product for 10 seconds or more, during this operation the device will power on and LED start flashing for 3 seconds.

Step3: After hold the magnet close to Hall switch duration more than 10 seconds, remove the magnet and keep it away from the device and then hold the magnet close to the device again within 2 seconds.

Step4: The LED on the device will keep solid red for 3 seconds, and then the device will automatically restart. After completing the above **steps**, you are done restoring factory Settings.

The following figure describes the flowchart about the hardware reset operation using button or magnets:

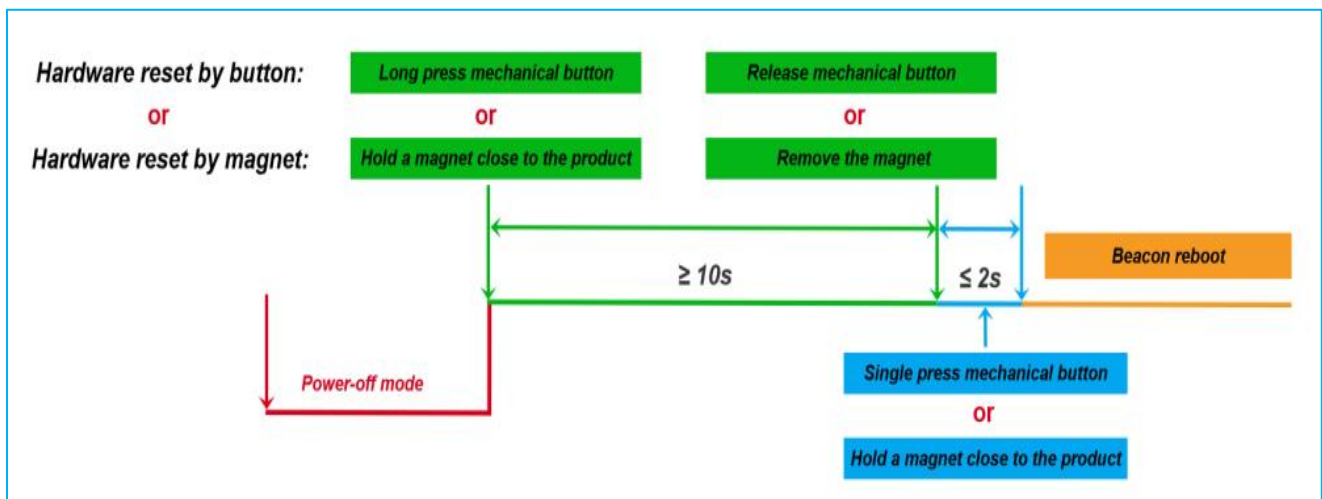


Figure 23: Hardware reset by button/magnet flowchart

➤ Hardware reset by battery:

In order to avoid the stuck in the dilemma of not being able to restore factory settings due to the hardware design without physical buttons, we have provided a universal operation flow to execute the hardware reset.

Step1: Remove the cover and the battery.

Step2: Put the battery in until you see the LED keep blinking for 2 seconds. This LED blinking means that the battery is connected successfully.

Step3: When the LED blinking finishes, put the battery in again within 5 seconds, and then repeat the process four more times, removing and replacing the battery. You should see a LED blinking each time you put the battery in, for a total of five blinks. The fifth blinking is different from the previous four and indicates that the device is factory reset successfully.

1st – 4th : LED keep blinking for 2s. This indicates the battery is connected successfully.

5th : LED keep solid for 2s. This indicates the device is factory reset successfully.

Step4: Replace the cover by aligning the battery room and press down on the cover.

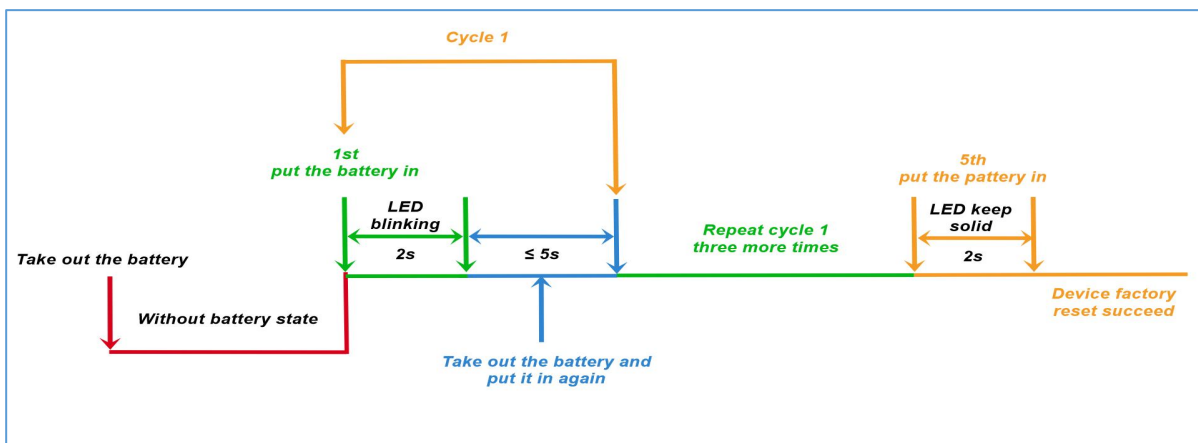


Figure 24: How to hardware reset you Tag

Remark:

1. Software reset won't reset connection password.

The hardware factory reset feature is enabled by default. If you don't need this function, you can disable it through the APP.

3.9 Modify "Connection password"

In MOKO firmware, password verification is ON by default and "Moko4321" was set as default connection password. You can modify your Beacon's connection password by just tapping **"SETTING >>> modify password"** option, and then input the password as you want.

Remark: Connection password should not exceed 16 characters in length.

3.10 Remote reminder

Enter this configuration item, where you can control the device's LED lights and buzzer. Please make sure that your device is equipped with LED or a buzzer before proceeding.

3.11 Reset Battery

The device supports broadcasting and displaying the remaining battery level as a percentage. If you have replaced the battery with a new one, please connect the device and enter “**SETTING>>>Reset Battery**” configuration item to reset the battery, ensuring that the device can calculate the battery level starting from 100%. Additionally, please make sure that the battery you replace is a new one with a full charge.

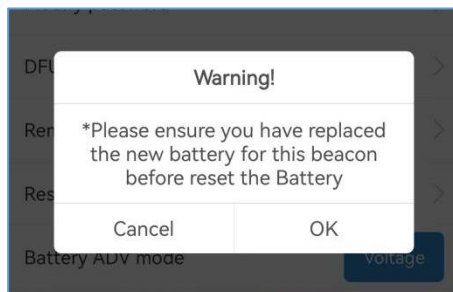


Figure 25: Reset Battery warning page

3.12 Battery ADV Mode

The device supports the choice of whether to broadcast battery voltage or battery percentage in the **Sensor info** frame, you can enter the “**SETTING>>>Battery ADV mode**” and choose to broadcast one of them in this configuration item, this mode is mainly related to the **byte23-byte24** in the **Sensor info** frame.

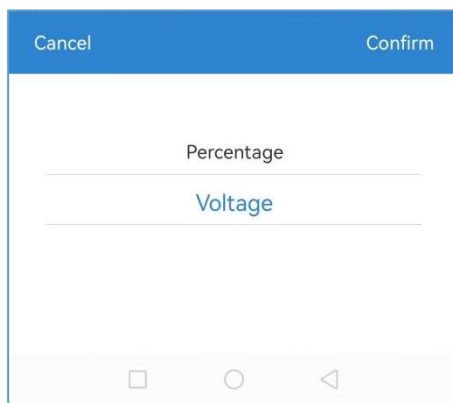


Figure 26: Battery ADV mode selection page

If you select the broadcast mode as Battery Voltage, then **byte23-byte24** inside the **Sensor info** broadcast frame will display the battery voltage, which is usually greater than 100. If you select the broadcast mode as Battery Percentage, then **byte23-byte24** inside the **Sensor info** broadcast frame will display the battery.

23-24	Battery percentage /Battery voltage	0x0C CC / 0x00 64	If the value is greater than 100, it represents the battery voltage. For example, 0x0CCC = 3276mV. If the value is less than or equal to 100, it represents the battery percentage. For example, 0x0064 = 100%.
-------	-------------------------------------	-------------------	--

When you parse the broadcast, you can use the above logic to determine which mode it belongs to(whether the value is greater than 100), so you can decide whether you should display the battery voltage or the battery percentage.

3.13 ADV Channel

This configuration item allows you to configure the BLE Bluetooth advertising channels. By default, the product broadcasts on BLE channels 37(2402MHz), 38(2426MHz), and 39(2480MHz) simultaneously. You can configure the device to broadcast on a single channel. The advantage of broadcasting on a single channel is lower power consumption, but the downside is that it will increase the difficulty for scanning devices to detect the broadcast.

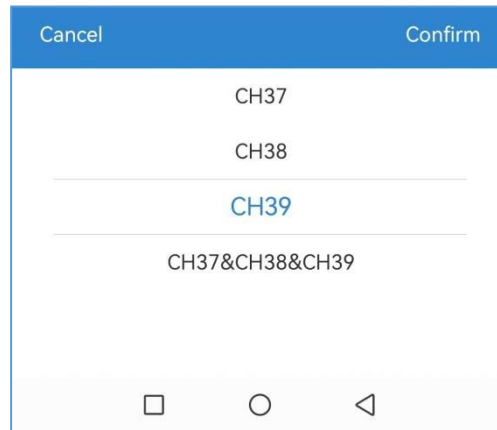


Figure 27: ADV channel selection page

3.14 How to enable “Long range” features

In order to enhance the transmission range of the tag and expand the application scenarios as well as reduce the gateway deployment costs, we have optimized the firmware feature and integrated with coded PHY mode. Based on current firmware, “1M PHY – Legacy” and “Coded PHY – Long Range” are supported already, and you can access to “**SETTING >>>ADV mode**” and adopt the right one to your application scenarios.

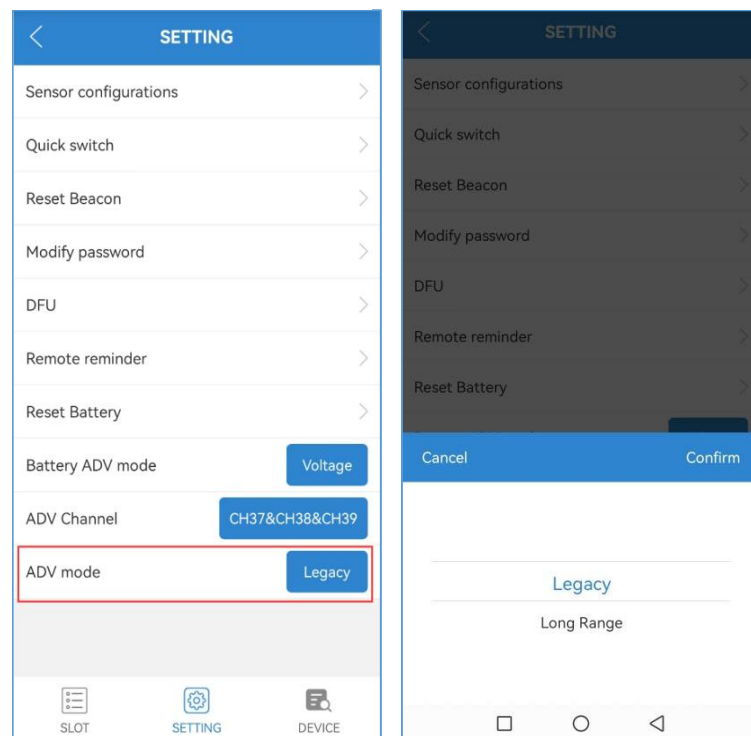


Figure 28: How to enable “Long Range” mode

➤ **Legacy** --- 1M PHY➤ **Long Rang** --- Coded PHY

Normally the transmission range in Coded PHY mode will be multiple times of transmission range in 1M PHY, and the advertisement payload format/content will be remaining with only the changes of advertisement type due to BLE 5.0 new specifications.

Here we have listed the ADV type difference between 1M PHY and Coded PHY. And for more introduction you can refer to link: <https://www.bluetooth.com/bluetooth-resources/intro-to-bluetooth-low-energy-coded-phy/>

Frame type	PHYs			
	LE 1M		LE Coded	
	Connectable status	Non-connectable status	Connectable status	Non-connectable status
iBeacon	ADV_IND + SCAN_RSP	ADV_SCAN_IND	ADV_EXT_IND + AUX_ADV_IND	
Eddystone – URL/TLM/UID	ADV_IND	ADV_NONCONN_IND	ADV_EXT_IND + AUX_ADV_IND	
Sensor info	ADV_IND + SCAN_RSP	ADV_SCAN_IND	ADV_EXT_IND + AUX_ADV_IND	

Table 10: ADV type difference between 1M and Coded PHY

3.15 Upgrade your Tag's firmware

The tag can realize OTA firmware update through DFU operations. Regarding of different import methods for android app and iOS app, here we will describe the detailed steps separately.

3.15.1 Android – MOKO APP DFU instructions

Step 1: Copy the upgrade package to a folder in the root directory of the smartphone.

Step 1: **Connect** with device and go to “**SETTING**” interface, press “**DFU**” option and then file manager route of smartphone will be prompted.

Step 2: Choose the required upgrade firmware package and the upgrade process will start automatically. Wait for the upgrade to complete, APP will disconnect with Beacon after upgrade completed successfully.

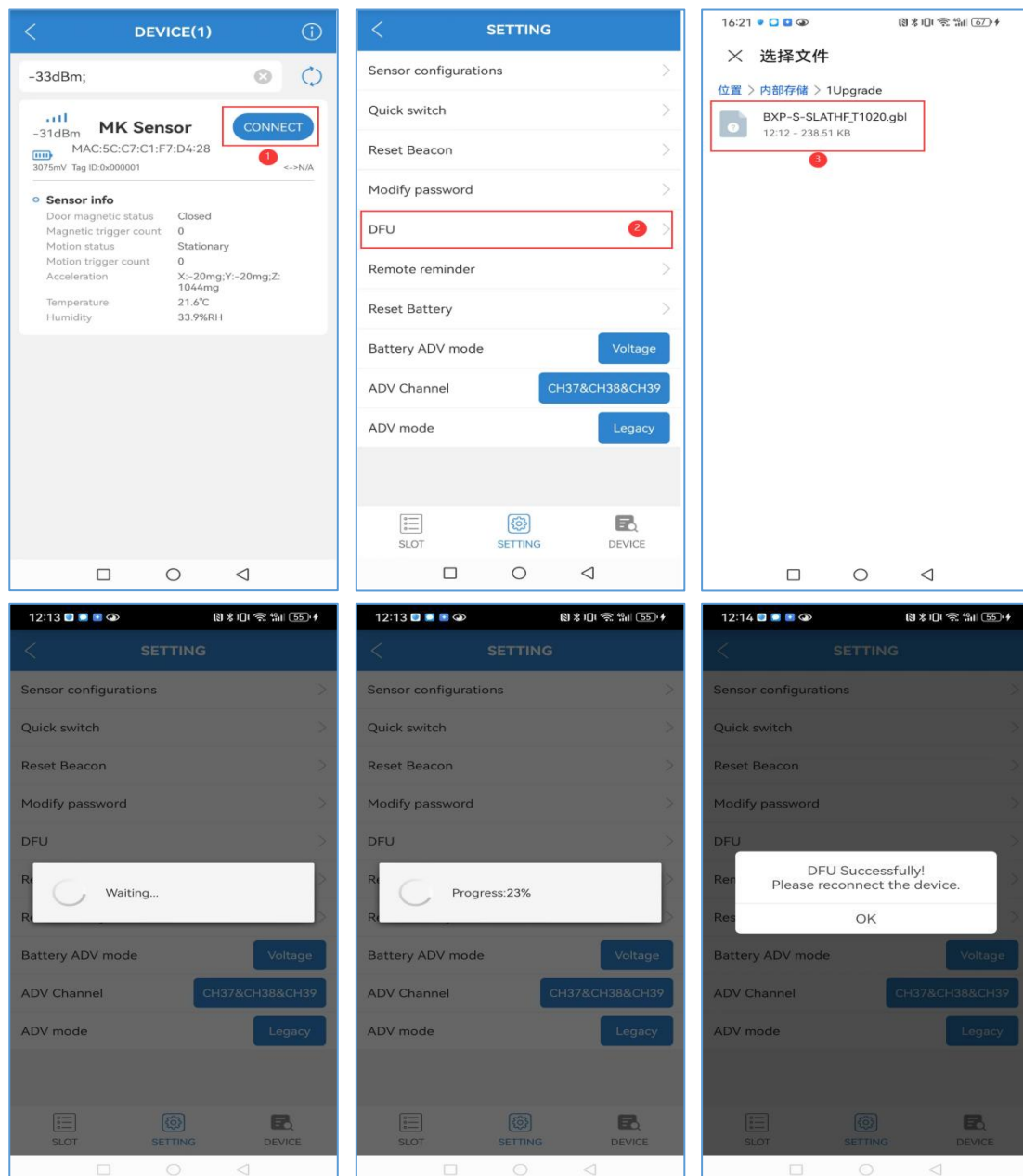


Figure 29: Firmware upgrade workflow (Android)

3.15.2 iOS – MOKO APP DFU instructions

Step 1: Install iTunes in PC and then import firmware upgrade file into “**BeaconX Pro**” APP, you can refer to below picture demonstrations – **Figure 35: Load upgrade package to nRF connect APP via iTunes.**

Step 2: **Connect** with device and go to “**SETTING**” interface, press “**DFU**” option and then select the upgrade file.

Step 3: The upgrade process will start automatically. Wait for the upgrade to complete, APP will disconnect with Beacon after upgrade completed successfully.

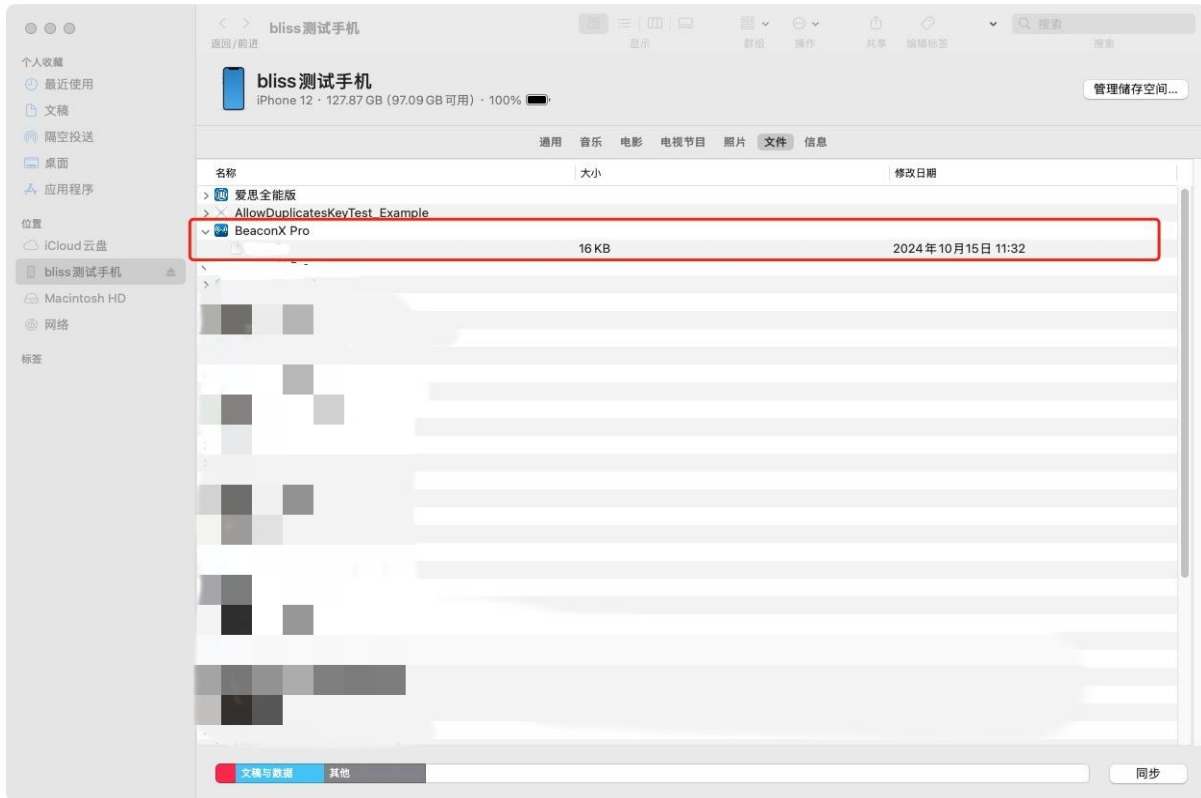


Figure 30: Load upgrade package to nRF connect APP via iTunes

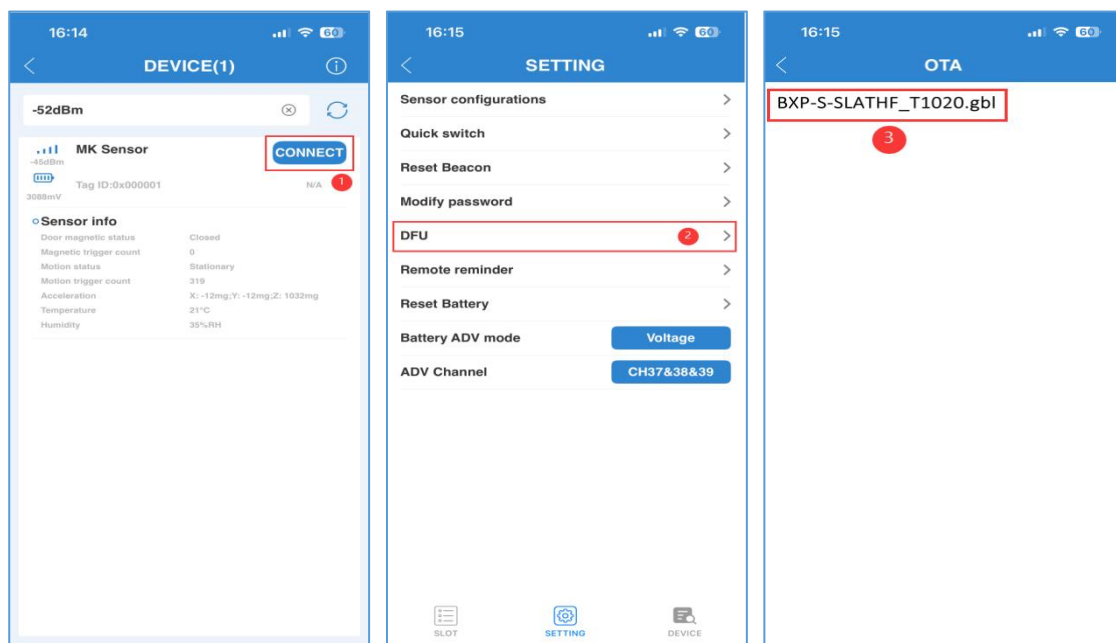


Figure 31: Firmware upgrade workflow of BeaconX Pro application (IOS)

3.16 General information

In some cases, you will need to know the firmware version of your Tag, or product model and etc. You can access to “**DEVICE**” interface to check and confirm the essential information.

DEVICE	
Battery voltage	3075mV
Battery Percentage	100%
Mac address	5C:C7:C1:F7:D4:28
Product model	MK Sensor
Software version	BXP-S-SLATHF
Firmware version	V1.0.2
Hardware version	MKBSL series
Manufacture date	2024/07/03
Manufacturer	MOKO TECHNOLOGY LTD.

Figure 32: Device information page

3.17 Obtain log file from Android APP

Sometimes when testing with our APP, there may occurred some issues that cannot be reproduced in MOKO side, then we will need you to help obtain the log file from android APP and then provide for us to further analysis.

Currently only Android APP support log record, and the **LOG** option is in “**About**” interface, you can export the log file and then send to us by following below steps if rare issue happened.

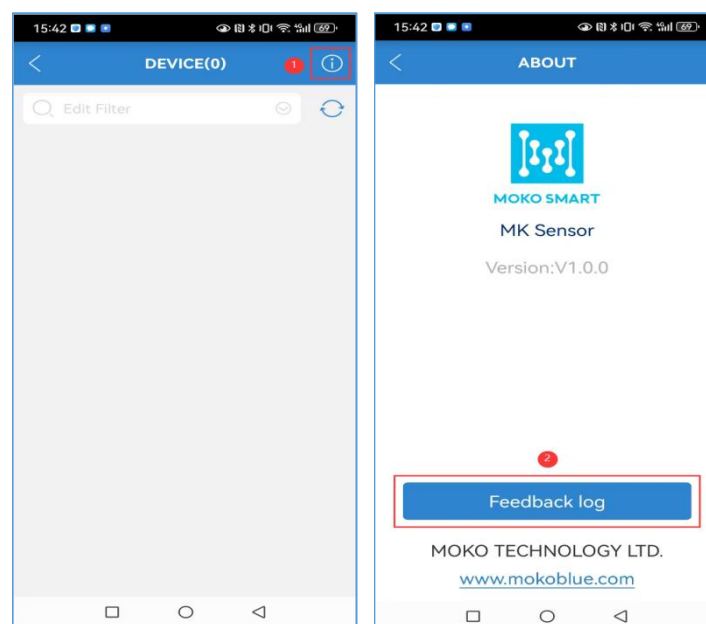


Figure 33: Obtain log file from Android APP

3.18 How to power off your device

The app does not support a software shutdown function because the firmware is compatible with different product models of the company, some of which do not have button or hall switch. If the device without button/hall switch is shut down via software, it cannot be powered on again, therefore, for devices without buttons, shutdown is not supported or recommended. For devices with buttons/hall switch, we recommend using the button/hall switch for shutdown.

The device's shutdown function via button/hall switch is disabled by default. You can refer to [Section 3.7.5 Disable/Enable your Beacon's button function](#) to connect to the device, enable the button shutdown function, and then disconnect. After that, you can shut down the device using the button/hall switch.

Additional Instructions:

The Tags with external button can power ON/OFF the device by button, such as S05T/L03/H4/H4P.

The Tags without external button but with Hall switches can be powered ON/OFF by Hall magnet, such as L01/L01S/L02/L02S/L04/M2 Sensor/M6.

The Tags without external button and without Hall switches do not support power OFF the device. such as M1P/M4/M4P/M4 Sensor

➤ ***Mechanical button:***

- **Power ON:** Long press button for more than 3 seconds, and LED will keep red blinking for 3 seconds.
- **Power OFF:** Long press button for more than 3 seconds, and LED will keep solid red for 3 seconds.

➤ ***Hall magnet:***

- **Power ON:** Hold a magnet close to the position marked with a Hall switch sign on the product and keep it there for 3 seconds , and LED will keep red blinking for 3 seconds.
- **Power OFF:** Hold a magnet close to the position marked with a Hall switch sign on the product and keep it there for 3 seconds and LED will keep solid red for 3 seconds.

NOTICE: For more details about operating procedures for each product, please contact the sales of MOKO SMART to get related Product Specifications documents.

4. FAQ

General Questions

- *What is RSSI?*

The RSSI (Received Signal Strength Indication) is a measurement of the power present in a received radio signal. It is usually displayed in dBm (Decibel-milliwatts). Low power devices, like Bluetooth Low Energy devices, are using very weak power to transfer data, therefore the usual values of RSSI are between -100dBm (Very weak signal) to around -20 dBm (strong signal).

- *Why the RSSI values are different on two Android devices.*

By definition the RSSI value is a measurement of the power present in received radio signal. The signal strength depends on the antennas, device orientation, location, distance from the phone to target device, humidity etc.

- *What's the difference between Advertising data and Scan Response data?*

In Bluetooth 4+ device, if not connected, continuously advertises with the Advertising data. When a scanner receives it, it may send the Scan request packet to obtain more advertising information. Then, the advertising device sends the Scan response data. Both packets may have up to 31 bytes.

APP Questions

- *Why the device name shows N/A?*

Device name is contained in scan response packet in “*Sensor info*” advertisement frame type, if “*Sensor info*” slot is not set, then APP will not achieve the device name and will show N/A accordingly.

Contact

MOKO TECHNOLOGY LTD.

An original manufacturer for IoT smart devices

Address: 4F, Building 2, Guanghui Technology Park, MinQing Rd, Longhua, Shenzhen, Guangdong, China

E-mail: Support_BLE@mokotechnology.com

Website: www.mokosmart.com

www.mokoblue.com